Membership vs. Pay-per-order: Delivery Platform Selection for Grocery Retailers

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Grocery retailers increasingly partner with third-party delivery platforms (e.g., Instacart, Shipt) to offer online delivery services. These platforms differ in how they charge customers for delivery: some use a *membership-based model*, charging a fixed fee for unlimited deliveries, whereas others use a *pay-per-order model*, charging per delivery. We conduct a multimethod study to examine how retailers should evaluate and choose among these platforms, combining a stylized analytical model with empirical evidence from a regional grocery chain that partners with both types of platforms across different geographic markets. Our analytical results show that platform partnerships do not necessarily generate economic benefits; rather their value depends on product characteristics, customer profiles, and platform design. Consistent with these insights; our empirical analysis finds that one partnership increases sales by 2.2%, whereas another decreases sales by 2.9%, with heterogeneous effects aligned with the model's predictions. To enhance partnership value, we posit that retailers should align their platform assortment with the platform's payment model. A counterfactual analysis suggests that a selective assortment–featuring more premium products on membership-based platforms and more staple products on pay-per-order platforms–can increase sales by up to 5.8% for the focal retailer.

Key words: Platform partnership; Online grocery delivery; Delivery payment model

1. Introduction

Customers increasingly rely on online channels for grocery shopping, with over 90% of U.S. shoppers engaging in both in-store and digital formats and digitally enabled grocery spending projected to reach $388 billion by 2027 NielsenIQ (2025). To remain competitive, many grocery retailers have expanded into in-house online delivery, but high fulfillment costs–driven by labor-intensive picking, packing, and last-mile delivery–strain already thin margins (Ghai 2024). Consequently, several retailers partner with third-party grocery delivery platforms that manage product display, in-store picking, and home delivery while pooling orders across multiple retailers to exploit economies of scale and reduce per-order fulfillment costs (Oberlo 2024). This rapidly growing market–projected to reach $1.12 trillion globally by 2028–includes both established grocery platforms such as Instacart and Shipt and multi-service delivery platforms such as DoorDash, Postmates, and Uber Eats (Meisenzahl 2024, Nasdaq 2024, YahooFinance 2024).

As this market continues to expand, an important yet understudied feature of retailer-platform partnerships is how platforms monetize delivery services. Grocery delivery platforms differ fundamentally in their primary payment mechanisms. For instance, Shipt (which partners with retailers such as Target and Publix in the U.S.) and Amazon Fresh (which partners with retailers such as Carrefour and Morrisons in Europe)[1] primarily employ a *membership-based model*, in which most customers pay a fixed monthly or annual fee and can place unlimited orders without per-order delivery charges. In contrast, Instacart and DoorDash (which partner with retailers such as Costco and Kroger in the U.S.) and Deliveroo (which partners with retailers such as Sainsbury's and Waitrose in Europe) have traditionally emphasized a *pay-per-order model*, in which most customers pay no membership fee but incur a delivery charge for each order. These delivery payment models likely induce distinct customer demand patterns. Membership-based platforms may stimulate customers to order more frequently to amortize the fixed subscription fee over more transactions. Conversely, pay-per-order platforms may reduce customer adoption frictions associated with upfront membership fees and expand the retailer's online market reach. Despite their strategic importance, it remains unclear how retailers should evaluate and prioritize these models when forming partnerships. Accordingly, we address the following research questions: *(i) What is the value of platform partnerships under membership-based versus pay-per-order models? and (ii) How should retailers choose between platforms primarily operating under membership-based versus pay-per-order models?*

To answer these questions, we obtain data from a grocery retailer operating 570 brick-and-mortar (BM) stores under four banners across seven U.S. states. In 2018, the retailer partnered with Shipt (primarily offering a membership-based model) for one banner and with Instacart (primarily using a pay-per-order model) for another banner. The retailer announced these partnerships only after launching platform-facilitated online delivery services, rendering the partnerships plausibly exogenous to customers' purchasing behavior and representing a unique opportunity for a quasi-experimental study. Leveraging this setting, we adopt a multimethod research approach. First, to derive theoretical insights, we develop a stylized analytical model capturing the key features of retail–platform partnerships and consumer purchasing behavior. Second, we empirically test the model's predictions using a difference-in-differences design on 39 weeks of product-level sales data before and after the partnerships.

Drawing on our analysis, we make several contributions. First, our theoretical model identifies a fundamental trade-off underlying retailer–platform partnerships, which operates through

[1] These examples reflect retail-platform partnerships as of April 2026.

two opposing mechanisms. On one hand, platform partnerships expand demand by enabling convenient home delivery and allowing customers to access the full assortment offered by all participating retailers. This attracts new customers who would otherwise be infeasible to serve by the BM store, increasing the retailer's sales. On the other hand, this same convenience motivates some existing BM customers–who would have completed their entire grocery shopping with the focal retailer in the absence of the platform–to switch to the platform and allocate their purchases to competing retailers. This reduces the retailer's share of customer spending. We refer to the first mechanism on new customers as the *market expansion effect* and the second on existing customers as the *market contraction effect*. The net impact depends on the balance between these two effects. Empirically, we find that partnering with Shipt increases retailer sales by 2.2% on average, while partnering with Instacart decreases sales by 2.9%. These results suggest that the market expansion effect dominates for Shipt, whereas the market contraction effect dominates for Instacart, highlighting that third-party grocery delivery partnerships can bring substantial benefits with some platforms but costly setbacks with others.

Second, our theoretical model characterizes that higher BM store prices or greater price dispersion across platform retailers amplify the value of platform partnership under the membership-based model, while diminishing that value under the pay-per-order model. In contrast, increases in customer consumption rates boost the value of platform partnership under both payment models. As BM store prices rise, fewer customers shop in-store, expanding the market the retailer can capture through the platform. Similarly, greater price dispersion introduces lower-priced alternatives on the platform, making it more attractive to new customers. At the same time, both higher BM prices and greater platform price dispersion encourage more existing BM customers to switch to the platform. This is where the membership-based model creates a key advantage. Under the pay-per-order model, switching remains easy due to the low per order delivery fee, so market contraction grows strongly relative to market expansion. Under the membership-based model, however, the high fixed membership fee acts as a substantial switching cost and makes switching harder. This limits customers switching and keeps market contraction in check, allowing market expansion to increasingly dominate. Lastly, higher consumption rates raise average cart values on platform orders for both new customers and those switching from the BM store. This boosts the market expansion effect while curbing the market contraction effect, collectively enhancing the value of platform partnership, regardless of the payment model. To test these findings empirically, we use in-store product price along with two proxy variables. To measure platform price dispersion, we obtained detailed pricing data from Numerator, covering all products offered by any retailer on Instacart and Shipt. For each product, we calculate the standard deviation of prices for similar products on a given platform. We operationalize the consumption rate of a product as the average number of units purchased per week across all customers buying that product. Consistent with our theoretical predictions, our empirical analysis shows that the effect of partnering with Shipt on sales is positively moderated by in-store price, platform price dispersion, and customer consumption rate. In contrast, the effect of partnering with Instacart on sales is negatively moderated by in-store price and platform price dispersion, but positively moderated by customer consumption rate.

Third, our theoretical model offers clear guidance on how retailers should choose between membership-based and pay-per-order platforms. The comparative statistics above immediately imply that as BM store prices or platform price dispersion increase, membership-based platforms become relatively more attractive. However, higher consumption rates raise the value of partnership under both payment models, making the comparison less straightforward. Our analysis shows that the relative advantage depends on consumption levels. When consumption is low, customers order infrequently, so the high fixed membership fee is costly relative to the small number of orders–making pay-per-order platforms more attractive and helping the retailer reach more customers. Conversely, when consumption is high, frequent orders spread the membership fee over many transactions, making membership-based platforms more valuable and enabling the retailer to attract more customers through this model. This non-monotonicity suggests that retailers operating in areas with low consumption rates (e.g., dense downtown areas with a higher prevalence of single-person households) are better suited to pay-per-order platforms, while retailers serving areas with high consumption rates (e.g., residential areas with larger households) benefit more from membership-based partnerships.

Finally, we show that retailers can further enhance platform partnership value by tailoring their platform assortment to the payment model. Higher-priced products with greater platform price dispersion are better suited for membership-based platforms, as they face lower market contraction and provide a stronger competitive advantage over rivals. The same benefits, however, can be achieved with lower-priced, low-dispersion products on pay-per-order platforms. We empirically quantify the gains from such selective assortment decisions in a counterfactual analysis. We find that limiting platform offerings to products identified by our theoretical model increases the sales impact of Shipt partnership from 2.2% to 8% and turns Instacart partnership from a negative 2.9% to a positive 1.4%. These results suggest that retailers should offer (i) premium or niche products with volatile prices on membership-based platforms and (ii) staple items with relatively steady prices on pay-per-order platforms.

Our findings have implications for grocery retailers deciding whether and how to partner with membership-based or pay-per-order platforms. In particular, the value of a partnership depends on products, customers, and platform characteristics, and can vary across different product offerings.

Overall, managers should carefully evaluate which platform partnership best aligns with these attributes and selectively determine which products to display through the platform to ensure alignment with their operational priorities.

2. Literature Review

Our study is related to three literature streams: third-party food delivery platforms, membership versus pay-per-order delivery services, and omnichannel retailing.

2.1. Third-Party Food Delivery Platforms

Much of the existing research in this stream has focused on the restaurant sector, documenting both the benefits and drawbacks of platform adoptions. For instance, Raj et al. (2020) highlight the critical role of food delivery platforms in sustaining small restaurant resilience when in-person dining was halted during the COVID-19 outbreak. Similarly, Li and Wang (2025) find that restaurants generally benefit from these platforms, with increases in both takeout and dine-in sales. However, prior research also documents important challenges. Karamshetty et al. (2020) show that heavy reliance on food delivery platforms hamper restaurants' demand forecasting accuracy and increases food waste. Leveraging an episode of these platforms' aggressive restaurant onboarding without permission, Mayya and Li (2021) find that platforms can lead to a net demand loss for independent restaurants, as declines in dine-in traffic outweigh gains in takeout orders. Addressing operational frictions between delivery and dine-in channels, Feldman et al. (2023) propose a coordinating contract to mitigate negative channel interactions. Finally, using a queuing model that distinguishes between traditional and tech-savvy customers, Chen et al. (2022) demonstrate that for restaurants with a large base of traditional customers, platform adoption may not be profitable, as platforms add delivery costs without sufficiently increasing demand.

Research on the adoption of food delivery platforms in the grocery sector remains relatively scarce. Unlike the restaurant industry, grocery retail is highly concentrated, with most platform-partnering retailers operating at national or regional scale. Consequently, grocery retailers may face distinct strategic and operational challenges when partnering with third-part delivery platforms. Using an analytical model, Delasay et al. (2021) examine the impact of the COVID-19 pandemic on grocery retail operations and show that platform adoption can increase store traffic, but may reduce retailer profits. Rabello de Castro (2020) study the timing of entry for a grocery delivery platform and highlight a first-mover advantage in building a loyal customer base, while Rabello de Castro (2019) analyze platform entry decisions in the presence of switching costs and show that switching costs significantly shape customer platform choice.

Despite this emerging literature, we are not aware of any study that examines third-party grocery delivery platform partnerships through the lens of platform payment structures. We contribute to this stream by (i) theoretically and empirically investigating the value of platform partnerships under membership-based versus pay-per-order models and (ii) analyzing how retailers should choose between these payment structures when forming platform partnerships.

2.2. Delivery Fee Structures in Online Retail

The literature on online delivery fee policies examines how different shipping fee structures affect consumer behavior and retailer profitability. Early work by Lewis (2006) and Lewis et al. (2006) shows that shipping fees significantly influence customer acquisition, retention, and purchase quantities. Gümüş et al. (2013) compare partitioned pricing–where shipping fees are charged separately–with non-partitioned pricing, in which delivery costs are included in the product price. Another stream of research studies contingent free shipping, where customers receive free delivery when their orders exceed a specified threshold. For example, Leng and Becerril-Arreola (2010) study joint pricing and contingent free shipping decisions in monopoly and duopoly settings; Hemmati et al. (2021) examine the interaction between contingent free shipping and return policies, and Li et al. (2023) investigate optimal thresholds for online grocery retailing, finding that consumer top-up behavior plays a key role in policy design.

More recently, research has focused on membership-based versus pay-per-order delivery. Balakrishnan et al. (2024) find that membership-based delivery increases profits by encouraging purchase frequency, while Belavina et al. (2017) show that memberships lead to more frequent but smaller grocery orders compared to pay-per-order pricing. Empirical work by Guo and Liu (2023) documents that membership-based delivery builds lock-in effects that increase purchase frequency and order size over time, and Fang et al. (2021) show that membership delivery enables better market segmentation, allowing retailers to charge higher product prices.

A key feature of prior research is its focus on retailer-controlled delivery fee policies. In contrast, platform-mediated delivery fee policies are determined by third-part platforms, requiring retailers to operate within platform-driven constraints. We contribute to this literature by examining how two types of platform-mediated delivery fee policies affect retail sales.

2.3. Omnichannel Retailing

Partnering with a third-party delivery platform constitutes a form of platform-enabled omnichannel retailing, whereby a brick-and-mortar retailer extends its physical operations into a digitally mediated ordering and delivery channel. A related stream of research has examined omnichannel partnerships that expand a retailer's service offerings in various settings. For example, Nageswaran et al. (2024) theoretically study a return partnership in which a pure online retailer collaborates with an offline retailer that serves as an in-store return location. Hwang et al. (2022) empirically demonstrate that return partnerships can generate incremental sales for the offline partner. Dan et al. (2022) analytically investigate showrooming partnership, focusing on how an online retailer selects between competing and non-competing offline partners. More recently, Jalali et al. (2026) examine a pickup partnership model in which an online retailer collaborates with an offline partner to provide in-store pickup for online orders. Despite these advances, this literature has thus far remained silent on partnerships involving third-party delivery platforms. Even within the broader omnichannel retailing literature, although prior studies have explored the integration of physical and digital channels through buy-online-pickup-in-store (Gallino and Moreno 2014, Gao and Su 2017), showrooming (Bell et al. 2018, Gao et al. 2022), and ship-to-store (Gallino et al. 2017, Ertekin et al. 2022), no study examines how offline and online channels interact when customers pay for online delivery under a membership-based model versus a pay-per-order model. We contribute to this stream by analyzing how delivery membership fees versus per-order delivery fees shape the dynamics between offline and online channels, when the online channel is operated by a third-party delivery platform.

3. Theoretical Model

In this section, we develop a stylized model to examine how platform partnerships affect retail sales under the membership-based and pay-per-order payment mechanisms, and to identify conditions guiding retailers' choice between the two. We first introduce the modeling framework. We then analyze the value of platform partnership under each model. Finally, we conduct comparative statics and derive theoretical predictions that we empirically test in Section 4.

3.1. Model Framework

Consider a grocery retailer with a brick-and-mortar (BM) store facing a decision on whether to partner with an online delivery platform. Consistent with the real-word practices described in the introduction, we examine two types of online delivery models. The first is a membership-based model, in which customers pay a fixed membership fee per unit time for unlimited free deliveries; we refer to this as the Online Membership platform, denoted by OM. The second is a pay-per-order model, in which customers pay a delivery fee for each order; we refer to this as the Online Pay-per-order platform, denoted by OP.

We consider customers who consume a product at a constant known rate r. When purchasing, they choose a quantity that is gradually consumed over time and replenish only once their inventory is depleted. To capture the trade-off between purchasing larger quantities (to reduce shopping frequency) and holding inventory at home, we assume that customers incur an inventory holding cost h per unit of grocery per unit time. This formulation reflects practical considerations such as limited storage space, perishability, or the cognitive burden of managing larger inventories. Customers' purchase quantities–whether from the BM store or, in the case of a partnership, from the online platform–as well as their channel choice are determined by the average costs associated with each channel. We summarize our notation in Table 1.

Table 1 List of notations

Parameter	Definition
$i \in \{\text{OM}, \text{OP}\}$	Index for platform partnership model: Membership ($i = \text{OM}$), and Pay-Per-Order ($i = \text{OP}$)
$v \sim U[0,1]$	Customer distance from the BM store; customers are heterogeneous w.r.t. v
h	Customer holding cost per unit per time
r	Customer consumption rate
p_b	Product price in BM store
p_o^i	Product price on online platform $i \in \{\text{OM}, \text{OP}\}$
μ_p	Mean price on an online platform
σ_p	Price dispersion on an online platform
$U(\mu_p - \sigma_p, \mu_p + \sigma_p)$	Price p distribution on an online platform with c.d.f. $F(p)$
p_K	Unit cost of purchasing from outside option (e.g. local convenience store)
ϕ	Fixed membership fee per unit time collected by the membership-based platform
λ	Delivery cost per order charged by the pay-per-order platform

3.2. Value of Online Platform

We model the value of partnering with an online delivery platform from the retailer's perspective, and quantify it through resulting change in customer demand. Specifically, we first derive the retailer's demand both without a partnership and with a partnership under each delivery model. We then compare the retailer demand under each platform partnership to the no-partnership benchmark to assess the value created by each type of platform partnership.

Demand without a Partnership. We begin by characterizing demand in the absence of a platform partnership. Following a Hotelling-type spatial framework, we assume that customers are uniformly distributed along a unit line, with distance from the BM store denoted by $v \in [0,1]$ (Belavina et al. 2017). To meet their consumption needs, customers visit the BM store and choose a purchase quantity Q_{BM} for each visit. Following (Ho et al. 1998), we model this decision using an economic order quantity (EOQ) tradeoff. Purchasing a larger quantity reduces store visit frequency–and thus travel costs–but increases inventory holding costs. Conversely, purchasing smaller quantities lowers holding costs while requiring more frequent store visits. The average cost per unit time for a customer located at distance v who purchases Q_{BM} per visit is

$$\text{Cost}(Q_{\text{BM}}) = p_b r + v\left(\frac{r}{Q_{\text{BM}}}\right) + h\left(\frac{Q_{\text{BM}}}{2}\right), \tag{1}$$

where p_b denotes the in-store price. The first term represents purchase expenditure. The second term captures the cost of store visits, where we normalize the unit travel cost to 1 without loss of generality. The third term reflects the average inventory holding cost. Minimizing $\text{Cost}(Q_{\text{BM}})$ with respect to Q_{BM} yields the optimal order quantity

$$Q^*_{\text{BM}} = \sqrt{\frac{2vr}{h}}.$$

Substituting back, the minimized average cost for a customer at distance v is

$$\text{Cost}_{\text{BM}}(v) = p_b r + \sqrt{2hvr}. \tag{2}$$

As expected, the customer's average cost increases in v at a decreasing rate. Let p_K denote the unit cost of outside option. To avoid trivial cases, we assume that $p_K \geq p_b$. The retailer's catchment area is defined as the maximal distance $v^{\text{catchment}}$ such that $\text{Cost}_{\text{BM}}(v) \leq p_K r$. Solving $\text{Cost}_{\text{BM}}(v) = p_K r$ yields

$$v^{\text{catchment}} = \frac{r(p_K - p_b)^2}{2h}. \tag{3}$$

Customers located in $[0, v^{\text{catchment}}]$ purchase from the retailer. Since customers are uniformly distributed, demand without a partnership is

$$D(\text{no partnership}) = v^{\text{catchment}} = \frac{r(p_K - p_b)^2}{2h}. \tag{4}$$

Demand with Platform Partnership. We now characterize demand under a platform partnership. In addition to purchasing from the BM store, customers can order through the online platform. While the platform features the retailer's product, it also displays the same product from competing retailers at different prices, creating competition among all participating sellers.

To model competition among retailers on the platform, we assume that platform prices are drawn from a distribution $F(p)$ defined over $[\mu_p - \sigma_p, \mu_p + \sigma_p]$, consistent with price-dispersion frameworks in Fibich et al. (2003), Amaldoss and He (2018), and Thaler (2008). For analytical tractability, we assume $F(p)$ is uniform, though the results extend to general continuous distributions. Let p_o^i denote the retailer's online price on platform i, where $i \in \{\text{OM}, \text{OP}\}$. The probability that a competing retailer offers a lower price is $F(p_o^i)$. Thus, a customer purchasing via the platform chooses the focal retailer with probability $1 - F(p_o^i)$ and a competitor with probability $F(p_o^i)$.

Let $\overline{v}^i$ represent the distance from the BM store of the customer who is indifferent between purchasing at the BM store and ordering through the online platform. Delegating the detailed analyses to Online Appendix A, we briefly outline how the threshold distance can be derived for each online partnership model.

- **OM model:** In this model, customers pay a fixed membership fee per unit time, denoted by ϕ, which covers unlimited free deliveries. Once the membership fee is paid, no additional delivery charges apply. Since customers order based on an *order-for-use* policy, which theoretically means $Q_{\text{OM}} \to 0$, holding cost under OM model becomes zero. Given that customers pay the lower of the retailer's online price p_o^i and a competing price p drawn from the distribution F, the total expected cost can be expressed as

$$\text{Cost}_{\text{OM}} = \phi + E_p \min(p_o^{\text{OM}}, p) r$$

where expectation is taken with respect to price distribution F in the online platform. Equating Cost_{OM} to $\text{Cost}_{\text{BM}}(v)$ and solving for $v \in [0,1]$ would yield the threshold $\overline{v}^{\text{OM}}$ as follows:

$$\overline{v}^{\text{OM}} = \frac{(\phi + r(E_p \min(p_o^{\text{OM}}, p) - p_b))^2}{2hr} \tag{5}$$

- **OP model:** In this model, customers pay a per-order delivery fee, denoted by λ. Because this introduces a fixed cost per order, the optimal purchase quantity under the OP model can be derived similar to the EOQ analysis. To avoid repetition, we only characterize the optimal purchase quantity on the platform by minimizing cost in the following EOQ expression:

$$\text{Cost}_{\text{OP}}(Q_{\text{OP}}) = E_p \min(p_o^{\text{OP}}, p) r + \lambda \left(\frac{r}{Q_{\text{OP}}}\right) + h \left(\frac{Q_{\text{OP}}}{2}\right) \tag{6}$$

which yields optimal $Q^*_{\text{OP}} = \sqrt{\frac{2\lambda r}{h}}$. Substituting this optimal order quantity into Equation (6), the overall cost of purchasing from the pay-per-order platform is given by

$$\text{Cost}_{\text{OP}} = E_p \min(p_o^{\text{OP}}, p) r + \sqrt{2h\lambda r}. \tag{7}$$

Equating Cost_{OP} to $\text{Cost}_{\text{BM}}(v)$ and solving for $v \in [0,1]$ yield the threshold $\overline{v}^{\text{OP}}$ as follows:

$$\overline{v}^{\text{OP}} = \frac{\left(\sqrt{2h\lambda r} + r(E_p \min(p_o^{\text{OM}}, p) - p_b)\right)^2}{2hr} \tag{8}$$

These threshold distance expressions give rise to the following customer segments:

- **Region** $[0, \overline{v}^i]$**:** Customers in this segment purchase from the BM store.
- **Region** $[\overline{v}^i, 1]$**:** Customers in this segment purchase through the online platform; the retailer retains them with probability $1 - F(p_o^i)$.

Aggregating across the regions, demand with platform partnership under model $i \in \{\text{OM}, \text{OP}\}$ can be expressed as follows:

$$D^i(\text{with partnership}) = \overline{v}^i + (1 - \overline{v}^i)(1 - F(p_o^i)). \tag{9}$$

Value of Platform Partnership. The value of platform partnership (VOP) can be expressed as the difference between demand with and without partnership:

$$\text{VOP}^i = D^i(\text{with partnership}) - D(\text{no partnership}).$$

By substituting the demand expressions with and without an online partnership–derived in Equations (9) and (4), respectively–we can characterize the conditions under which a platform partnership is value-enhancing.

PROPOSITION 1. *The value of partnership with online delivery platform $i \in \{\text{OM}, \text{OP}\}$ is as follows:*

$$VOP^i = \underbrace{(1 - v^{catchment})(1 - F(p_o^i))}_{\text{Market expansion}} - \underbrace{(v^{catchment} - \overline{v}^i)F(p_o^i)}_{\text{Market contraction}}. \tag{10}$$

where (i) $v^{catchment}$ is characterized in Equation (3) and (ii) $\overline{v}^i$ is characterized in Equations (5) and (8) for the OM *Model and the* OP *Model, respectively.*

The first term captures the incremental demand from customers who are outside the retailer's physical catchment area under no partnership, but purchase through the online platform once the partnership is established. We term this increase the *market expansion effect*. The second term reflects customers who previously purchased from the BM store under no partnership, but switch to buying from the platform after the partnership. Because some of their purchases may be diverted to competitors offering better prices, the retailer experiences a loss of sales from these customers. We term this decrease the *market contraction effect*. The overall VOP depends on the balance between these two opposing forces.

3.3. Sensitivity Analysis

Proposition 1 indicates that an online delivery platform partnership does not necessarily create value for the retailer. This raises a practical question: under what conditions should a retailer expect the partnership to be beneficial versus detrimental? To address this, we extend the managerial implications of our model through a sensitivity analysis. Specifically, we examine how the VOP varies with (i) the customer consumption rate r, (ii) the in-store price p_b, and (iii) the online platform price dispersion σ_p. These parameters capture key features of customers, the retailer's product, and the platform environment, respectively, and are typically observable or estimable by retailers.

Recall from Proposition 1 that VOP^i reflects the net effect of market expansion and market contraction. Accordingly, the comparative statics depend on how model parameters influence these two opposing effects.

PROPOSITION 2. *There exist platform-specific price cutoffs $\bar{p}_b^{\text{OM}}$ and $\bar{p}_b^{\text{OP}}$ such that, if the BM-store price satisfies $\bar{p}_b^{\text{OM}} \le p_b \le \bar{p}_b^{\text{OP}}$, then* VOP$^{\text{OM}}$ *is increasing in r, p_b, and σ_p, whereas* VOP$^{\text{OP}}$ *is increasing in r but decreasing in p_b and σ_p, where*

$$\bar{p}_b^i \equiv p_o^i - \frac{\left(\sigma_p - (\mu_p - p_o^i)\right)^2}{4\sigma_p}, \qquad i \in \{\text{OM}, \text{OP}\}.$$

Proposition 2 presents the sensitivity analysis of the VOP for moderate in-store prices, which are likely representative of our empirical setting. Results for other price levels are provided in Online Appendix A.

Customer consumption rate (r). For both platform types, a higher consumption rate increases VOP. Intuitively, higher consumption rates increase both the frequency and the average cart value of platform orders. Under OM, more frequent orders allow customers to spread the fixed membership fee ϕ over a large number of purchases, effectively lowering the per-purchase costs. Under OP, larger cart values make it easier for customers to justify the per-order fee λ. Consequently, the convenience of the online delivery platform increases, which enhances the market expansion effect and mitigates the market contraction effect in Equation (10).

In-store price (p_b). Holding online platform-specific price p_o^i fixed, changes in store price p_b affect VOP through the switching thresholds $v^{\text{catchment}}$ and $\overline{v}^{\,i}$. Using the decomposition in Proposition 1, we can rewrite

$$\text{VOP}^i = \big(1 - F(p_o^i)\big) - v^{\text{catchment}} + F(p_o^i)\,\overline{v}^{i}, \qquad i \in \{\text{OM}, \text{OP}\}.$$

Differentiating with respect to p_b yields

$$\frac{\partial \text{VOP}^i}{\partial p_b} = \underbrace{-\frac{\partial v^{\text{catchment}}}{\partial p_b}}_{\text{effect on market expansion}} + \underbrace{F(p_o^i)\,\frac{\partial \overline{v}^i}{\partial p_b}}_{\text{effect on market contraction}}, \qquad i \in \{\text{OM}, \text{OP}\}. \tag{11}$$

Since $v^{\text{catchment}} = \frac{r(p_K - p_b)^2}{2h}$, we have

$$\frac{\partial v^{\text{catchment}}}{\partial p_b} = -\frac{r(p_K - p_b)}{h} \leq 0.$$

Thus, a higher p_b expands the mass of customers captured by the online partnership. This implies that the market expansion effect in the VOP expression increases in p_b. At the same time, a higher p_b decreases $\overline{v}^i$, which in turn increases the share of customers who switch from the BM store to online and are exposed to competitors' products on the online marketplace. This amplifies the market contraction component. Consequently, the overall comparative static effect of p_b depends on which force—market expansion or market contraction—dominates.

In order to answer this, we need to investigate how threshold expression $\overline{v}^i$ changes with respect to p_b for each model $i \in \{\text{OM}, \text{OP}\}$. From the threshold expressions in Equations (5) and (8), we obtain the following:

$$\frac{\partial \overline{v}^{\text{OM}}}{\partial p_b} = -\frac{\phi + r(E_p \min(p_o^{\text{OM}}, p) - p_b)}{h}, \qquad \frac{\partial \overline{v}^{\text{OP}}}{\partial p_b} = -\frac{\sqrt{2h\lambda r} + r(E_p \min(p_o^{\text{OP}}, p) - p_b)}{h}.$$

These expressions show that the two platforms differ in the magnitude of the market contraction term: under OP, the threshold $\overline{v}^{\text{OP}}$ contains the additional component $\sqrt{2h\lambda r}$ associated with the per-order fee, which is generally larger than the corresponding component ϕ in OM that reflects the fixed membership fee. Consequently, the market contraction effect under OP becomes more sensitive to increases in p_b through $\overline{v}^{\text{OP}}$ relative to the market expansion effect. In contrast, under OM, the market contraction component grows more moderately with p_b, allowing the market expansion effect to dominate. These results imply that the VOP^{OM} increases and VOP^{OP} decreases in p_b as summarized in Proposition 2.

Platform price dispersion (σ_p). Finally, platform price dispersion influences the VOP in a manner similar to p_b. Higher price dispersion increases the likelihood that customers encounter lower prices on the platform, making both platform types more attractive. This strengthens the market expansion effect but also encourages customers to switch from the BM store to the platform, amplifying the market contraction effect. The magnitude of this contraction differs across platforms: it is substantial under OP, driven by the component $\sqrt{2h\lambda r}$, and moderate under OM, governed by the component ϕ. Consequently, VOP^{OM} increases while VOP^{OP} decreases with higher platform price dispersion.

3.4. Which Delivery Platform to Work with?

In Section 3.2, we show that partnering with an online delivery platform can create value for the retailer only when the resulting market expansion effect outweighs the market contraction effect. The remaining strategic question is therefore comparative: when both partnerships are viable, which model creates greater value—a membership-based platform or a pay-per-order platform? To answer this, we compare the VOP under the two partnerships and define

$$\Delta_{\text{VOP}} \equiv \text{VOP}^{\text{OP}} - \text{VOP}^{\text{OM}}, \tag{12}$$

which measures the additional (or reduced) value the retailer derives from partnering with OP relative to OM.

The platform choice hinges on a simple yet fundamental tradeoff: *customer consumption rate*. This rate determines whether customers will use the online platform repeatedly. Under OM, customers pay a fixed membership fee ϕ and face zero marginal delivery cost, so the value of delivery scales with purchase frequency. In contrast, under OP, customers incur a per-order fee λ, so delivery convenience is evaluated on a transaction-by-transaction basis, making it easier to adopt occasionally, but costly to use repeatedly.

The following proposition characterizes the conditions under which each model creates more value for the retailer by analyzing how the Δ_{VOP} changes with respect to system parameters:

PROPOSITION 3. *The additional value the retailer derives from partnering with* OP *relative to* OM *(Δ_{VOP}) is decreasing in p_b, σ_p, and r.*

Proposition 3 can be interpreted as a practical *guideline* centered on two key questions. The first is: *how competitive is our online offer on each platform?* Changes in p_b or σ_p do not simply make the online platforms uniformly more or less attractive; rather, they reshape the balance between market expansion and contraction. This balance, in turn, depends on whether customers are likely to use delivery repeatedly. While higher p_b or σ_p may enhance the relative attractiveness of online platforms, they also intensify leakage, amplifying the shift from the BM store to the platform as a contraction effect. Here, the platform-specific delivery payment mechanism becomes critical. Under OM, the fixed fee acts as a switching cost that discourages migration, thereby keeping market contraction in check. Consequently, the OM platform generates greater value than OP at high levels of p_b or σ_p, whereas the opposite holds at lower levels. This highlights that platform choice should not be framed in terms of which platform attracts more traffic, but rather in terms of which one more effectively limits leakage from the BM store to the platform.

The second question is: *are we serving in a low- or high-frequency replenishment environment?* Higher consumption rates imply more frequent purchases, making it easier for customers to amortize the membership fee over time. In contrast, when consumption is low, the fixed fee becomes as a barrier, and the per-order fee is easier to justify given infrequent use. As a result, the OM platform creates greater value than OP at high levels of r, whereas the reverse holds at low levels of r.

3.5. List of Hypotheses

In the next section, we present empirical evidence supporting our theoretical model. Specifically, to complement our analyses in sections 3.2 and 3.3–which focus on the value of platform partnership and sensitivity analyses, respectively–we empirically test the theoretical predictions summarized in Table 2.

Table 2 Theoretical Predictions Derived from the Analytical Model

Hypothesis	Prediction	Analytical support
$H_{1a/1b}$	OM platform partnership increases/decreases sales.	Proposition 1
$H_{2a/2b}$	OP platform partnership increases/decreases sales.	Proposition 1
H_3	The effect of OM platform partnership on sales is positively moderated by (a) in-store price, (b) customer consumption rate, and (c) platform price dispersion.	Proposition 2
H_4	The effect of OP platform partnership on sales is (a) negatively moderated by in-store price, (b) positively moderated by customer consumption rate, and (c) negatively moderated by platform price dispersion.	Proposition 2
H_5	The moderating effects of (a) in-store price, (b) customer consumption rate, and (c) platform price dispersion on the sales effect of platform partnerships will be more positive for OM-based platforms than for OP-based platforms.	Proposition 3

4. Empirical Application

In this section, we use retail data to empirically evaluate the predictions derived from our theoretical model. We begin by describing the empirical setting and data, followed by our identification strategy and the estimation results.

4.1. Empirical Setting and Data

We obtained data from a grocery retailer that operates 570 stores under four banners across seven southern U.S. states: Alabama, Florida, Georgia, Louisiana, Mississippi, North Carolina, and South Carolina. The retailer holds a strong regional position, accounting for approximately 40% of the grocery market in its operating areas (Data Axle 2018). Its customer base is highly local, with over 50% of customers making purchases at least every other week. Decision-making is centralized at the corporate level, encompassing both operational decisions such as inventory and pricing and strategic decisions including store openings/closures and external partnerships, to ensure alignment with the retailer's long-term objectives and overall brand strategy.

Prior to May 2018, all banners operated exclusively through brick-and-mortar (BM) stores. In May 2018, the retailer introduced same-day online delivery for two banners, partnering with Shipt for one banner (46 stores) and Instacart for another banner (245 stores). Online delivery was launched simultaneously for all these stores on May 7, 2018. The remaining 279 stores under the other two banners remained BM-only throughout the study period. Partnership decisions were made at the corporate level, and platform assignments were dictated by exclusive regional availability—each banner's operating region was served by only one platform. This created clear geographic separation across the three groups of stores. The retailer announced these partnerships only after May 7, 2018, making the introduction of same-day online delivery plausibly exogenous to customer behavior.

Customers with access to online delivery could create an account on Shipt or Instacart via the platforms' app or website, browse the retailer's full assortment, add items to a virtual cart, choose a delivery window, and receive their order at their doorsteps. Orders were fulfilled by platform-employed shoppers (i.e., drivers). The platforms independently managed all aspects of online transactions, including setting platform-specific prices, processing payments, and dispatching shoppers to pick items in-store and deliver them to customers. The retailer, however, earned revenue as if the purchase occurred by the customers in-store: items were picked by shoppers directly from store shelves and processed through the standard in-store checkout system, so the retailer received the in-store price for each unit sold, regardless of the price charged to customers by the platforms. No additional contractual arrangements governed these transactions.

Both platforms applied price premiums over retailers' in-store prices as platform revenue (Lampariello 2023). However, they differed in their pricing models, fee structures, and retail scale, with Shipt generally charging lower premiums (Savings Lifestyle 2023) but partnering with fewer retailers than Instacart (Contrary Research 2023, Redman 2024). Shipt operated under a strictly membership-based model: customers were required to purchase an annual $99 membership to access the service, which included unlimited free delivery on orders exceeding $35. Without a membership, the platform was entirely inaccessible as it did not provide a per-order option for non-members. In contrast, Instacart, primarily followed a pay-per-order model, charging delivery fees that varied by location and time of day, typically ranging from $3.99 to $7.99 per order. Although Instacart also offered an optional Express membership, only a small fraction of customers subscribed, and even these users were still required to a per-order charge in the form of a mandatory 5% service fee on every order. Despite these differences, the two platforms were comparable in delivery performance. Both offered similar services (i.e., same-day with scheduled time windows), relied on independent shoppers, and matched shoppers to orders based on their real-time proximity to the store and the customer's delivery address.

We use item-level transaction data spanning 39 weeks–from January 1 to September 30, 2018–including 19 weeks before and 20 weeks after the launch of the platform partnership. The data cover 570 stores across all four banners. Each transaction record is associated with a unique identifier and contains information on price, wholesale cost, quantity, product description, hierarchical product taxonomy (i.e., department, category, subcategory), purchase time, and store location, with products identified by unique stock keeping units (SKUs). The dataset also includes customer identifiers derived from loyalty card usage, covering approximately 95% of customers.

Consistent with our theoretical model, we analyze outcomes at SKU level and define the primary main unit of analysis as a SKU-store-week triplet (indexed by i, s, and t, respectively). We aggregate transactions at this level to construct weekly sales for each product in each store. Our main sample consists of 1,392 SKUs that were continuously available in all 570 stores over the study period. These SKUs span seven departments: dairy, dry grocery, frozen goods, general merchandise, non-food grocery, packaged meat, and beverages.

4.2. Mapping the Theoretical Model to the Data

Our theoretical model features three key parameters–in-store price (p_b), customer consumption rate (r), and platform price dispersion (σ_p)–which jointly determine the impact of partnership on retailer sales. Below, we describe how we operationalize these constructs in the data.

Sales: We operationalize sales ($Sales_{ist}$) as the natural logarithm of net sales for product i in store s during week t. Net sales are defined as weekly sales net of returns and wholesales costs (i.e., the procurement prices paid by the retailer to suppliers).

In-store price: We define the in-store price ($Price_{is}$) as the natural logarithm of the average retail price of product i in store s during the pre-platform partnership period.

Customer consumption rate: To construct this variable, for product i in store s, we first identify all unique customers who purchased the product in that store. For each customer, we identify the weeks of their first and last purchases during the pre-platform partnership period and treat this interval as the period during which the customer was active for product i in store s. We then compute the customer's purchase rate as the total number of units purchased divided by the number of active weeks. Finally, we operationalize customer consumption rate for product i in store s ($Consumption_{is}$) as the natural logarithm of the average of these individual purchase rates across all customers. We assume that the consumption rates estimated from purchases by 95% of customers (i.e., loyalty card users) are representative of the entire customer base.

Platform price dispersion: Recall that this construct captures the price dispersion of similar products on a platform–information not available in the retailer's own transaction data. To construct this variable, we obtain pricing data from Numerator–a market research firm that collects multichannel transaction records from a representative panel of U.S. customers–for all products listed on Shipt and Instacart between May 7 and September 30, 2018. For each product, the dataset includes unit price, quantity purchased, purchase date, retailer name, delivery platform (Instacart or Shipt), product description, and hierarchical product taxonomy. Because Instacart and Shipt use different product descriptions and hierarchical taxonomies, which also differ from the retailer's own labeling, we implement a text-based similarity matching procedure to link products across datasets (see Online Appendix B). Specifically, for each product i in the retailer's transaction data, we identify similar products on both platforms within the Numerator dataset, restricting matches to the same department (e.g., meat, dairy) to ensure meaningful comparability. We then measure the price dispersion of product i as the standard deviation of prices among its matched counterparts across retailers on a given platform. This yields two product-level dispersion measures–$Dispersion_i^S$ and $Dispersion_i^I$–corresponding to the Shipt and Instacart platforms, respectively.

4.3. Identification Strategy

Our goal is to estimate the causal effect of launching online delivery through a partnership with a third-party platform on retail sales. We implement a difference-in-differences (DiD) approach that compares the post-platform partnership change in per-product sales after May 2018 at stores that partnered with Shipt or Instacart (i.e., *treatment group*) with that at stores that remained BM-only during the study period (i.e., *control group*). This strategy provides two key identification advantages while accounting for baseline differences between groups. First, by examining within-group changes over time, it differences out any time-invariant unobserved heterogeneity that could otherwise bias estimates (Angrist and Pischke 2009). Second, by comparing these changes across groups, it adjusts for common time trends in sales, isolating the effect attributable to the introduction of online delivery services.

Given the longitudinal structure of our data and Hausman test results rejecting the random-effects specification, we estimate the DiD model using a fixed-effects panel regression framework, specified as follows:

$$Sales_{ist} = \beta_0 + \beta_1 Treatment_s \times After_t + \mathbf{X}_{is}^{'}\beta_c + \mu_{is} + \tau_t + \epsilon_{ist}, \tag{13}$$

where (i) $Treatment_s$ is an indicator equal to one if store s belongs to the treatment group and zero otherwise; (ii) $After_t$ equals one for observations after May 7, 2018 and zero otherwise; (iii) $\mathbf{X}_{is}$ is a vector of time-invariant controls, including $Price_{is}$, $Consumption_{is}$, and $Dispersion_i$; (iv) (μ_{is} denotes SKU-store fixed effects; (v) τ_t denotes week fixed effects; and (vi) ϵ_{ist} is the random error term. Because of the fixed-effects specification, the SKU-store fixed effects absorb the coefficients on time-invariant variables in $\mathbf{X}_{is}$. The DiD coefficient β_1 captures the effect of launching online delivery via a third-party platform on sales for treatment stores, corresponding to the value of platform partnerships in the theoretical model, and is used to test Hypotheses 1 and 2. We report robust standard errors clustered at the product–store level.

To assess how the effect of launching online delivery via a third-party platform on sales is moderated by in-store price, customer consumption rate, and platform price dispersion (i.e., Hypotheses 3 and 4), we specify the following fixed-effects triple-differences model:

$$\begin{aligned} Sales_{ist} = {} & \beta_0 + \beta_1 Treatment_s \times After_t + \beta_2 Price_{is} \times Treatment_s \times After_t \\ & + \beta_3 Price_{is} \times After_t \times + \beta_4 Consumption_{is} \times Treatment_s \times After_t \\ & + \beta_5 Consumption_{is} \times After_t + \beta_6 Dispersion_i \times Treatment_s \times After_t \\ & + \beta_7 Dispersion_i \times After_t + \mathbf{X}_{is}^{'}\beta_c + \mu_{is} + \tau_t + \epsilon_{ist}, \end{aligned} \tag{14}$$

The coefficients of interest to test Hypotheses 3 and 4 are β_2, β_4, and β_6 for in-store price, customer consumption rate, and platform price dispersion, respectively.

Because the retailer made partnership decision at the banner level based on platform availability in each banner's operating region, treatment and control stores may differ systematically in observable characteristics. To improve comparability, we implement propensity score matching using store-level covariates measured in the pre-platform partnership period. These covariates include (i) the number of SKUs per store, capturing assortment breadth and store size; (ii) the average number of checkout registers, capturing labor and sales floor capacity (Friebel et al.

2022); (iii) the number of households and the average household income in the store's ZIP code, capturing local market conditions; and (iv) the average product price in the store's assortment, capturing assortment value. The matching procedure, described in Online Appendix C, yields 168 control stores successfully matched to 42 treatment stores offering online delivery via Shipt and 226 treatment stores offering online delivery via Instacart. The resulting matched sample comprises 7,176,667 observations for the Shipt partnership analysis and 13,452,754 observations for the Instacart partnership analysis. Table 3 provides the descriptive statistics of the matched sample.

Table 3 **Descriptive Statistics**

	Partnership with Shipt		Partnership with Instacart		No Partnership	
Variables	Mean	Std.Dev	Mean	Std.Dev	Mean	Std.Dev
$Sales$	1.677	1.023	1.793	1.069	1.791	1.079
$Price$	0.865	0.602	0.899	0.618	0.897	0.610
$Consumption$	-0.432	0.501	-0.318	0.452	-0.323	0.450
$Dispersion^S$	0.736	0.367	–	–	–	–
$Dispersion^I$	–	–	0.767	0.379	–	–
Sample Size	1,416,569		7,692,656		5,760,098	

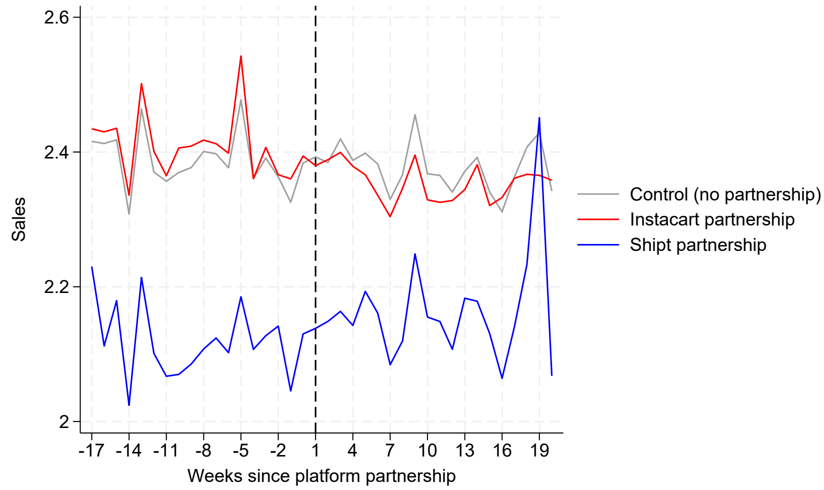
The DiD model relies on the parallel trends assumption, which implies that in the absence of partnership, the trend in sales should be the same between treatment and control groups. Using the matched data, we assess the parallel trends assumption in two ways. First, we plot the average weekly SKU sales for control groups and treatment groups in Figure 1. The vertical line marks the launch of online delivery via third-party delivery platforms. Consistent with the parallel trends assumption, we observe that prior to the launch, sales trends appear similar across all three groups of stores. Following the launch, compared to the control group stores, treatment stores that partnered with Shipt show a slight increase in sales, while treatment stores that partnered with Instacart exhibit a decline in sales.

Second, to formally examine these patterns during the pre-platform partnership period, we estimate the following regression using only pre-platform partnership observations:

$$Sales_{ist} = \beta_0 + \beta_1 Treatment_s \times Trend_t + \mathbf{X}_{is}^{'}\beta_c + \mu_{is} + \epsilon_{ist} \tag{15}$$

where $Trend_t$ denotes the number of weeks since the beginning of the data for an observation. The coefficient β_1 captures the difference in sales trends between treatment and control groups, with a

Figure 1 **Average weekly sales across matched stores**



statistically significant β_1 indicating a violation of the parallel trends assumption. Since we have two treatment groups, we estimate Equation (15) separately for each group, using the same control group in both cases. Table 4 presents the results. Column (1) reports results for treatment stores that partnered with Shipt and control stores, and Column (2) for treatment stores that partnered with Instacart and control stores. In both cases, β_1 is not statistically significant, providing evidence in support of the parallel trends assumption during the pre-platform partnership period.

Overall, the parallel trends analysis, together with propensity score matching, validates the DiD identification strategy, and we next present results from the DiD model estimations.

Table 4 **Parallel Trends Assumption Test**

Variables	(1) Partnership with Shipt	(2) Partnership with Instacart
Treatment×Trend	-0.0003	0.0002
	(0.0009)	(0.0005)
SKU-Store FE	Yes	Yes
Week FE	Yes	Yes
Observations	3,396,821	6,373,783
R-squared	0.687	0.688

Note: (1) Clustered robust standard errors are reported in parentheses,
*** p<0.001, ** p<0.01, * p<0.05, + p<0.1

4.4. Results

We estimate Equations (13) and (14) separately for each platform partnership and report the results in Table 5 for Shipt and Table 6 for Instacart. In each table, Column(1) presents estimates from Equation (13), Column (2) reports estimated from Equation (14) including only the in-store price moderator, Column (3) additionally incorporates the customer consumption rate moderator, and Column (4) reports the full specification with all three moderators. We assess the significance of adding an additional moderator in a model using a Wald test between the model with the additional moderator and the model without the additional moderator. As suggested by the Wald statistics, as we sequentially move from Column (1) to Column (4), model fit improves significantly at each step. Therefore, we use Column (4) to test Hypothesis 3 for both partnerships.

For the Shipt partnership reported in Table 5, the DiD coefficient is positive and statistically significant ($\beta_1 = 0.022$ in Column (1)). This indicates that partnering with a delivery platform employing a membership-based model increases sales by 2.2%, supporting Hypothesis 1a. The triple difference estimates in Column (4) are also positive and statistically significant ($\beta_2 = 0.032$, $\beta_4 = 0.017$, and $\beta_6 = 0.010$). These results indicate that the sales-enhancing effect of a membership-based platform partnership is stronger for products with higher in-store prices, higher customer consumption rates, and greater platform price dispersion, providing support for Hypothesis 3 in the Shipt context.

Table 5 Estimation Results for the Shipt Partnership

Variables	(1)	(2)	(3)	(4)
Treatment×After	0.022***	-0.006*	-0.005	-0.009**
	(0.002)	(0.003)	(0.003)	(0.004)
Price×Treatment×After		0.031***	0.034***	0.032***
		(0.004)	(0.004)	(0.004)
Price×After		-0.031***	-0.037***	-0.041***
		(0.001)	(0.002)	(0.002)
Consumption×Treatment×After			0.017***	0.017***
			(0.004)	(0.004)
Consumption×After			-0.027***	-0.028***
			(0.002)	(0.002)
Dispersion×Treatment×After				0.010*
				(0.005)
Dispersion×After				0.027**
				(0.002)
SKU-Store FE	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes
Observations	7,176,667	7,176,667	7,176,667	7,176,667
R-squared	0.667	0.667	0.667	0.667
Wald Stat.	-	209.83***	102.84***	105.60***

Note: (1) Clustered robust standard errors are reported in parentheses,
*** p<0.001, ** p<0.01, * p<0.05, + p<0.1

Turning to the Instacart partnersip in Table 6, in contrast to the Shipt result, the DiD coefficient is negative and statistically significant ($\beta_1 = -0.029$ in Column (1)). This indicates that partnering with a delivery platform employing a pay-per-order model reduces sales by 2.9%, providing support for Hypothesis 2b. The triple difference estimates in Column (4) are negative and statistically significant for the in-store price and platform price dispersion moderators ($\beta_2 = -0.020$ and $\beta_6 = -0.008$), but positive and statistically significant for the customer consumption rate moderator ($\beta_4 = 0.008$). These findings indicate that the sales-reducing effect of a pay-per-order platform partnership is exacerbated for higher (in-store) priced products and those subject to greater platform price dispersion, but alleviated for products with higher customer consumption rates. Overall, these results align with the theoretical predictions for pay-per-order delivery platform in Proposition 2, supporting Hypothesis 4.

Table 6 Estimation Results for Instacart Partnership

Variables	(1)	(2)	(3)	(4)
Treatment×After	-0.029***	-0.008***	-0.007***	-0.004*
	(0.001)	(0.002)	(0.002)	(0.002)
Price×Treatment× After		-0.023***	-0.021***	-0.020***
		(0.002)	(0.002)	(0.002)
Price×After		-0.031***	-0.037***	-0.042***
		(0.001)	(0.002)	(0.002)
Consumption×Treatment× After			0.008***	0.008**
			(0.002)	(0.002)
Consumption× After			-0.027***	-0.028***
			(0.002)	(0.002)
Dispersion×Treatment× After				-0.008**
				(0.003)
Dispersion×After				0.028***
				(0.002)
SKU-Store FE	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes
Observations	13,452,754	13,452,754	13,452,754	13,452,754
R-squared	0.663	0.663	0.663	0.663
Wald Stat.	-	1189.94***	173.36***	163.17***

Note: (1) Clustered robust standard errors are reported in parentheses,
*** $p<0.001$, ** $p<0.01$, * $p<0.05$, + $p<0.1$

To summarize, our analyses reveal contrasting effects of delivery platform partnerships in our setting. Partnering with a membership-based platform like Shipt increases sales, particularly for higher-priced products with higher customer consumption rates and greater platform price dispersion. In contrast, partnering with a pay-per-order platform like Instacart reduces sales, with losses being larger for higher-priced products and those subject to greater platform price dispersion, though the negative impact is mitigated for products with high customer consumption.

4.5. Comparing Partnership Value between OM and OP Platforms

In Section 3.4, our theoretical model characterizes how a retailer should choose between a membership-based platform and a pay-per-order platform when forming a partnership with a third-party delivery platform. The model further predicts that retailers are more likely to partner with a membership-based platform, rather than a pay-per-order platform, as their assortments feature higher-priced products, higher customer consumption rates, and greater platform price dispersion. We formalize this prediction as Hypothesis 5. In this section, we provide empirical evidence to test this hypothesis.

We begin by pooling the sample from both treatment groups with the control group. We then replace the binary treatment indicator, $Treatment_s$, with a categorical variable capturing store-level treatment assignment, where $Treatment_s = 0$ for control stores, $Treatment_s = S$ for stores that partnered with Shipt, and $Treatment_s = I$ for stores that partnered with Instacart. We further replace $Dispersion_i$ with $Dispersion_i^S$ and $Dispersion_i^I$, which denote platform price dispersion for Shipt and Instacart, respectively. Using the combined sample, we re-estimate Equations 13 and 14 with these modified variables, and report the results in Table 7: Column (1) presents estimates from Equation (13) and Column (2) presents estimates from Equation (14).

We find that the DiD coefficients from the pooled sample in Column (1) are consistent with those reported in Tables 5 and 6, providing a sanity check for this approach. To test Hypothesis 5, we perform a Wald test for each moderator, comparing the estimated triple difference coefficients in Column (2) across the two platform partnerships. The results indicate that the moderating effect is more positive for the Shipt partnership than for the Instacart partnership with respect to in-store price (Wald $\chi^2 = 583.64$, $p < 0.001$ for $\beta = 0.031$ vs. $\beta = -0.020$), customer consumption rate (Wald $\chi^2 = 10.64$, $p < 0.01$ for $\beta = 0.017$ vs. $\beta = 0.008$), and platform price dispersion (Wald $\chi^2 = 42.85$, $p < 0.001$ for $\beta = 0.014$ vs. $\beta = -0.006$). These findings imply that as in-store prices, customer consumption rates, and platform price dispersion increase, the sales impact of a platform partnership is more favorable for a membership-based platform like Shipt than for a pay-per-order platform like Instacart, supporting Hypothesis 5.

4.6. Selective Platform Assortment

Our theoretical result in Proposition 3, supported by the empirical evidence in Section 4.5, suggests that retailers can enhance the value of platform partnerships by tailoring their online assortment: (i) high-priced products with high consumption rates and high platform price dispersion are better suited for membership-based platforms, while (ii) low-priced products with low consumption rates and low platform price dispersion are better suited for pay-per-order platforms. In this subsection, we empirically examine this alternative assortment strategy and its potential to improve the outcomes of platform partnerships. Specifically, we conduct a counterfactual analysis to estimate the sales that would have occurred had the retailer partnered with the two platforms using a selective product assortment, as predicted by our theory.

Table 7 Estimation Results from Pooled DiD Models

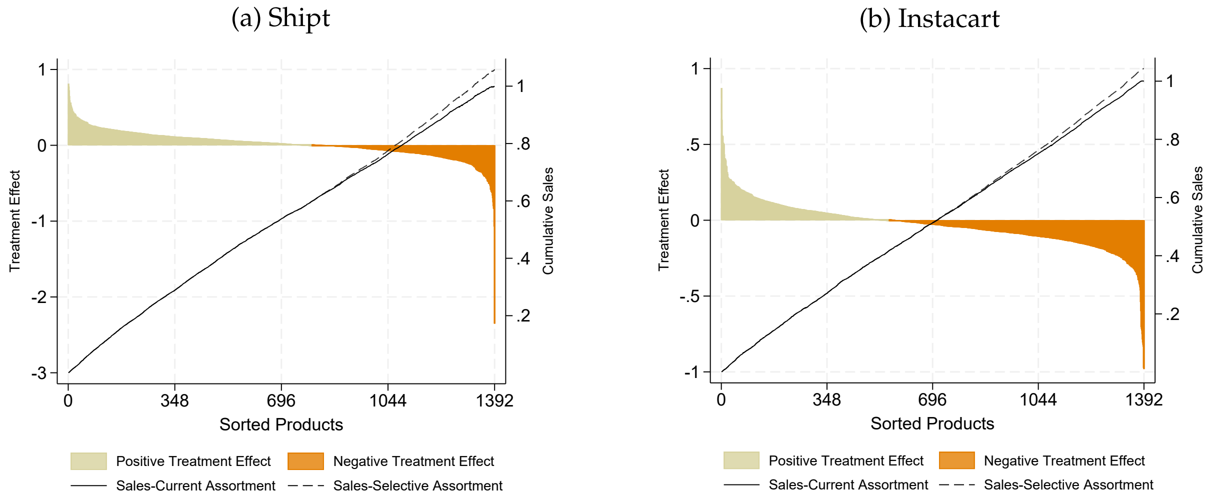
Variables	(1)	(2)
[S]Treatment×After	0.022***	-0.010***
	(0.002)	(0.002)
[I]Treatment×After	-0.029***	-0.005**
	(0.001)	(0.001)
Price×[S]Treatment×After		0.031***
		(0.002)
Price×[I]Treatment×After		-0.020***
		(0.001)
Consumption×[S]Treatment×After		0.017***
		(0.003)
Consumption×[I]Treatment×After		0.008***
		(0.002)
Dispersion^S × [S]Treatment×After		0.014***
		(0.003)
Dispersion^I × [I]Treatment×After		-0.006***
		(0.002)
SKU-Store FE	Yes	Yes
Week FE	Yes	Yes
Observations	14,869,323	14,869,323
R-squared	0.661	0.661

Note: (1) Clustered robust standard errors are reported in parentheses.
(2) For brevity, the coefficients of After×Price, After×Consumption, After×Dispersion^{Shipt}, and After×Dispersion^{Insta} are omitted from the table.
*** p<0.001, ** p<0.01, * p<0.05, + p<0.1

To identify products to include in the selective assortment, we extend our empirical framework (see Online Appendix D) to estimate the sales impact of launching online delivery via a third-party platform partnership for each of the 1,392 products. Figure 2 summarizes the product-level treatment effects for each platform partnership (Shipt on the left and Instacart on the right). In each panel, the bar plot depicts product-level treatment effects for products in the treatment group: the sand-shaded area represents products with positive estimated treatment effects, while the orange-shaded area represents products with negative estimated treatment effects. For the Shipt partnership, the product-level treatment effect is positive for 57.18% of products in treatment stores (i.e., 796 products in the sand-shaded area) and negative for the remaining 42.82% (i.e., 596 products in the orange-shaded area). For the Instacart partnership, we estimate that 39.73% of products have positive treatment effects (i.e., 499 products in the sand-shaded area), whereas 60.27% have negative treatment effects (i.e., 839 products in the orange-shaded area).

These findings suggest that the retailer could have improved the overall sales impact of platform partnerships by restricting the platform assortment to products in the sand-shaded areas. To quantify this improvement, as detailed in Online Appendix D, we conduct a counterfactual analysis in which we predict post-intervention product-level sales under (i) the current strategy, where all 1,392 products are offered on both platforms, and (ii) the proposed selective assortment strategy, where only products in the sand-shaded areas are offered on the respective platforms.

Figure 2 **Effect of Selective Assortment on Sales**



We then compute each product's percent contribution to total sales and plot the cumulative sales distribution in Figure 2. In each panel, the dotted line denotes cumulative sales under the current strategy, while the dashed line denotes cumulative sales under the selective assortment strategy. We find that a selective platform assortment could increase overall sales by 5.8% for the Shipt partnership, raising its sales impact from 2.2% to 8%, and by 4.32% for the Instacart partnership, shifting its sales impact from -2.9% to +1.4%.

5. Conclusion

Establishing partnerships with online delivery platforms has become a strategic necessity for brick-and-mortar grocery retailers seeking to remain competitive in an increasingly digital marketplace. However, as our analysis demonstrates, the value of such partnerships depends critically on the delivery payment structure adopted by the platform. In this paper, we develop a unified analytical and empirical framework to evaluate two dominant models—membership-based and pay-per-order delivery—and show that their impact on retailer performance is governed by a fundamental trade-off between market expansion and market contraction due to within-platform competition.

Our results reveal that these two forces operate in opposite directions and are shaped by product, customers, and platform characteristics. While platform partnerships expand the retailer's geographic reach and attract new demand, they also intensify competition by enabling existing customers to switch to competing retailers on the same platform. The net value of a partnership therefore hinges on which of these forces dominates. Empirically, we find that partnering with a membership-based platform increases sales (approximately +2.2%), whereas partnering with a pay-per-order platform decreases sales (approximately -2.9%). These contrasting effects highlight that platform partnerships are not universally beneficial; rather, their success depends on aligning platform design with the retailer's demand environment.

Beyond average effects, our analysis uncovers several dimensions of heterogeneity that provide direct guidance for managerial decision-making. First, customer consumption rate plays a central role in determining the relative attractiveness of delivery formats. In high-consumption environments—such as suburban areas or households with frequent replenishment needs—the membership-based model is more effective, as the fixed fee is amortized over many transactions and reduces per-order frictions. In contrast, in low-consumption environments—such as dense urban areas with smaller households—the pay-per-order model lowers adoption barriers and can be more effective in attracting marginal customers. This suggests that retailers should not view platform choice as a one-size-fits-all decision, but rather as a function of local demand characteristics.

Second, in-store price levels and price dispersion critically shape the competitive dynamics on the platform. For higher-priced products or categories with greater cross-retailer price dispersion, membership-based platforms are more advantageous because they limit market contraction by introducing a switching cost through the membership fee. In contrast, pay-per-order platforms exacerbate market contraction in such environments, as customers can easily compare and switch to lower-priced alternatives. For staple products with low prices and limited dispersion, however, pay-per-order models can remain competitive due to their lower entry barriers. This insight implies that retailers should align delivery formats not only with customer segments but also with product categories.

Third, our counterfactual analysis demonstrates that selective platform assortment is a powerful lever for improving partnership performance. Rather than offering the full assortment on all platforms, retailers can significantly enhance outcomes by tailoring product offerings to the strengths of each delivery model. Specifically, allocating high-margin or high-dispersion products to membership-based platforms and limiting pay-per-order platforms to commodity-like products can substantially increase the value of both partnerships—transforming negative outcomes into positive ones in some cases. This finding highlights that platform strategy should extend beyond partner selection to include assortment design and channel differentiation. From this perspective, our results support retailers such as Walmart, which offer a more limited and curated assortment on delivery platforms (e.g., Instacart).

Overall, our findings suggest that retailers should move beyond the simplistic question of whether to partner with a delivery platform, and instead focus on how to structure and manage such partnerships. The value of platform partnership depends not only on access to online demand, but also on how platform design interacts with consumer behavior, product characteristics, and competitive dynamics. By explicitly accounting for these interactions, retailers can make more informed partnership decisions and design more effective omnichannel strategies.

Several avenues for future research remain. First, while our baseline analysis focuses on settings in which each platform employs a single delivery model, the industry is increasingly shifting toward hybrid structures that combine membership and pay-per-order options. Our extended analysis (Online Appendix E) suggests that the fundamental trade-off between market expansion and market contraction identified in our paper continues to hold in hybrid structures. It also suggests that, in such environments, customers self-select into the delivery payment option that minimizes their total cost, with consumption rate serving as the key sorting mechanism. This trend raises important strategic questions. For example, future research could examine how retailers should position themselves within multi-modal platforms and influence customer migration across channels through pricing, assortment, and promotions. Likewise, it would be valuable to study how third-party delivery platforms should optimally balance these two payment mechanisms within a unified service offering. Second, incorporating richer behavioral considerations—such as loyalty formation, search frictions, and platform-driven promotions—could further refine our understanding of substitution dynamics. Finally, extending the analysis to other retail contexts and platform-mediated services would help generalize the insights and broaden their applicability.

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Online Supplement: Membership vs. Pay-per-order: Delivery Platform Selection for Grocery Retailers

A. Proofs

In order to prove Proposition 1, we need following technical Lemmas.

LEMMA 1. *Suppose the retailer operates solely through the BM store. The customer chooses to purchase from BM store if $v \in [0, v^{catchment}]$, and outside option if $v \in [v^{catchment}, 1]$, where $v^{catchment} = \frac{r(p_K - p_b)^2}{2h}$.*

Proof of Lemma 1. Without a partnership, the grocery store sells only through the BM store. Customers choose between two purchasing options: (i) purchasing from an outside option at a total cost of $p_K r$, or (ii) visiting the BM store, incurring a cost of $v \sim U[0,1]$, and purchasing $Q^*_{\text{BM}} = \sqrt{\frac{2vr}{h}}$, resulting in the total cost of $C_{\text{BM}} = p_b r + \sqrt{2hvr}$. By comparing the total cost under both scenarios, one can verify that the shopper chooses to purchase from the BM store if $v \in [0, v^{\text{catchment}}]$, and outside option if $v \in [v^{\text{catchment}}, 1]$, where $v^{\text{catchment}} = \frac{r(p_K - p_b)^2}{2h}$. ■

LEMMA 2. *Let $\phi^{\text{OM}} = \frac{\left(\sigma_p^2 - 2(\mu_p + p_o^{\text{OM}})\sigma_p + (\mu_p - p_o^{\text{OM}})^2\right)r + 4p_K r \sigma_p}{4\sigma_p}$. When partnering with* OM*-based platform, the customer's optimal purchasing decision is as follows*

- *If $\phi \leq \phi^{\text{OM}}$: the customer chooses to purchase from the BM store if $v \in [0, \overline{v}^{\text{OM}}]$, and online with membership option if $v \in [\overline{v}^{\text{OM}}, 1]$, where $\overline{v}^{\text{OM}} = \frac{\left(\phi + r(E_p \min(p_o^{\text{OM}}, p) - p_b)\right)^2}{2hr}$.*
- *If $\phi > \phi^{\text{OM}}$: the customer chooses to purchase from the BM store if $v \in [0, v^{catchment}]$, and outside option if $v \in [v^{catchment}, 1]$, where $v^{catchment} = \frac{r(p_K - p_b)^2}{2h}$.*

Proof of Lemma 2. To find the customer's optimal purchasing decision, we need to compare its overall cost from three different options. As discussed earlier, the overall cost of shopping from an outside option (i.e., local convenience store) is $p_K r$. The overall cost by shopping from the BM store is $C_{\text{BM}} = p_b r + \sqrt{2hvr}$. Finally, if the customer chooses to purchase online form the membership-based platform, their total cost, including membership fee and inventory cost, would be $\text{Cost}_{\text{OM}} = \phi + E_p \min(p_o^i, p) r$, where

$$E_p \min(p_o^i, p) = p_o^i - F(p_o^i)\left(\frac{p_o^i - \mu_p + \sigma_p}{2}\right), \qquad i \in \{\text{OM}, \text{OP}\}. \tag{A.1}$$

By comparing cost of shopping from online platform Cost_{OM} to outside option $p_K r$, one can verify that the customer prefers order from online delivery platform to purchase from outside option when $\phi \leq \phi^{\text{OM}}$ where $\phi^{\text{OM}} = \frac{\left(\sigma_p^2 - 2(\mu_p + p_o^{\text{OM}})\sigma_p + (\mu_p - p_o^{\text{OM}})^2\right)r + 4p_K r \sigma_p}{4\sigma_p}$. Therefore, when $\phi > \phi^{\text{OM}}$, the customer compares the outside option to purchasing from the BM store, resulting in the same outcome as in Lemma 1. However, when $\phi \leq \phi^{\text{OM}}$, the customer compares the total cost of visiting the BM store with total cost of purchasing online form the membership platform. Considering the cost of visiting the BM store $v \sim U[0,1]$, the customer chooses purchasing from the BM store when $v \leq \overline{v}^{\text{OM}}$, where $\overline{v}^{\text{OM}} = \frac{\left(\left(\sigma_p^2-2(\mu_p-2p_b+p_o^{\text{OM}})\sigma_p+(\mu_p-p_o^{\text{OM}})^2\right)r-4\phi\sigma_p\right)^2}{32hr\sigma_p^2} \equiv \frac{\left(\phi+r(E_p\min(p_o^{\text{OM}},p)-p_b)\right)^2}{2hr}$. ■

LEMMA 3. *Let* $\lambda^{\text{OP}} = \frac{\left(\left(\sigma_p^2-2(\mu_p+p_o^{\text{OP}})\sigma_p+(\mu_p-p_o^{\text{OP}})^2\right)r+4p_Kr\sigma_p\right)^2}{32hr\sigma_p^2}$. *The customer's optimal purchasing decision is as follows*

- *If* $\lambda \leq \lambda^{\text{OP}}$: *the customer chooses to purchase from the BM store if* $v \in [0, v^{\text{BM}\to\text{OP}}]$, *and online from the pay-per-order platform if* $v \in [v^{\text{BM}\to\text{OP}}, 1]$, *where* $\overline{v}^{\text{OP}} = \frac{\left(\sqrt{2h\lambda r}+r(E_p\min(p_o^{\text{OM}},p)-p_b)\right)^2}{2hr}$.
- *If* $\lambda > \lambda^{\text{OP}}$: *the customer chooses to purchase from the BM store if* $v \in [0, v^{catchment}]$, *and outside option if* $v \in [v^{catchment}, 1]$, *where* $v^{catchment} = \frac{r(p_K-p_b)^2}{2h}$.

Proof of Lemma 3. Similar to proof of Lemma 2, we need to compare the customer's overall cost from three different options: First, the overall cost of shopping from an outside option (i.e., local convenience store) is $p_K r$. Second, the overall cost by shopping from the BM store is $C_{\text{BM}} = p_b r + \sqrt{2hvr}$. Finally, if the customer opts to purchase online from the pay-per-order platform, then the total cost is given by

$$\min_{Q_{\text{OP}}} \text{Cost}_{\text{OP}} = E_p \min(p_o^{\text{OP}}, p) r + \lambda \left(\frac{r}{Q_{\text{OP}}}\right) + h\left(\frac{Q_{\text{OP}}}{2}\right) \tag{A.2}$$

By plugging the optimal order quantity $Q_{\text{OP}} = \sqrt{\frac{2\lambda r}{h}}$ into Eq. (A.2), one can verify that the overall cost of purchasing online frmo the pay-per-order platform is as follows:

$$C_{\text{OP}} = E_p \min(p_o^{\text{OP}}, p) r + \sqrt{2h\lambda r} \tag{A.3}$$

By comparing cost of shopping from online platform in Eq. (A.3) to outside option $p_K r$, one can verify that the customer prefers online order to outside option when $\lambda \leq \lambda^{\text{OP}}$ where $\lambda^{\text{OP}} = \frac{\left(\left(\sigma_p^2-2(\mu_p+p_o^{\text{OP}})\sigma_p+(\mu_p-p_o^{\text{OP}})^2\right)r+4p_Kr\sigma_p\right)^2}{32hr\sigma_p^2}$. Therefore, when $\lambda > \lambda^{\text{OP}}$, the customer compares the outside option to purchasing from the BM store, resulting in the same outcome as in Lemma 1. However, when $\lambda \leq \lambda^{\text{OP}}$, the customer compares the total cost of visiting the BM store with total cost of purchasing online from the pay-per-order platform (i.e., Eq. (A.3)). Considering the cost of visiting BM store $v \sim U[0,1]$, the customer chooses purchasing from the BM store when $v \leq \overline{v}^{\text{OP}}$, where

$$\overline{v}^{\text{OP}} = \frac{\left(\sigma_p^2 - 2(\mu_p - 2p_b + p_o^{\text{OP}})\sigma_p + (\mu_p - p_o^{\text{OP}})^2\right)^2 r - 8\sigma_p\left(\sigma_p^2 - 2(\mu_p - 2p_b + p_o^{\text{OP}})\sigma_p + (\mu_p - p_o^{\text{OP}})^2\right)\sqrt{2h\lambda r} + 32h\lambda\sigma_p^2}{32hr\sigma_p^2} \tag{A.4}$$

Finally, one can verify that the above equation is equivalent to $\overline{v}^{\text{OP}} = \frac{\left(\sqrt{2h\lambda r}+r(E_p\min(p_o^{\text{OM}},p)-p_b)\right)^2}{2hr}$.

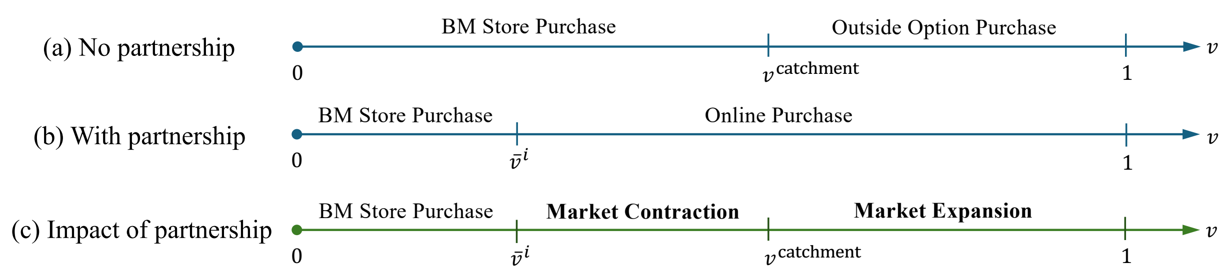
■

Proof of Proposition 1. The value of partnership can be characterized by comparing the grocery store demand without (Lemma 1) and with partnership (Lemmas 2 and 3). In what follows, we elaborate on each case separately.

Recall from Lemma 1 that, without partnership, the customer chooses to purchase from BM store if $v \in [0, v^{\text{catchment}}]$, and outside option if $v \in [v^{\text{catchment}}, 1]$, where $v^{\text{catchment}} = \frac{r(p_K - p_b)^2}{2h}$. Figure (A.1-a) presents demand characterization without partnership.

With partnership, the customer has three choices: (i) shop from an outside option, (ii) shop from the BM store, and (iii) shop from an online delivery platform (either membership or pay-per-order platform). Clearly, when shopping from outside option is more attractive than online purchases, customers will not choose to buy online. In such cases, partnering with an online delivery platform brings no value to the grocery store. For the membership-based platform, this situation arises when the membership fee is high (i.e., $\phi > \phi^{\text{OM}}$). For the pay-per-order platform, it occurs when the delivery cost is relatively high (i.e., $\lambda > \lambda^{\text{OP}}$). However, when the membership fee is low (i.e., $\phi \le \phi^{\text{OM}}$ in the case of partnering with a membership-based platform), or delivery cost is low (i.e., $\lambda \le \lambda^{\text{OP}}$ in the case of partnering with a pay-per-order platform), then online purchasing becomes more attractive than the outside option. Under this condition, as discussed in Lemmas 2 and 3, customers choose between shopping at the BM store or online as shown in Figure (A.1-b).

Figure A.1 Impact of Partnership on Demand Characterization



By comparing Figures (A.1-a) and (A.1-b) one can characterize the impact of partnership with platform $i \in \{\text{OM}, \text{OP}\}$ on demand characterization, as shown in Figure (A.1-c). Three distinct segments would appear: (i) When $v \in [0, \overline{v}^i], i \in \{\text{OM}, \text{OP}\}$, customers who would choose the BM store without a partnership continue to do so, meaning the partnership has no impact. (ii) When $v \in [\overline{v}^i, v^{\text{catchment}}], i \in \{\text{OM}, \text{OP}\}$, customers who would choose the BM store without a partnership decide to purchase from the online delivery platform. This results in *market contraction* because customers may end up buying a similar product from a competitor in the online channel, which occurs with probability $F(p_o)$. (iii) When $v \in [v^{\text{catchment}}, 1]$, customers who would choose the outside option without a partnership now pay the membership fee (in the case of membership platform) or pay for delivery (in the case of pay-per-order platform) and purchase from the online delivery platform. This leads to a *market expansion* with probability $1 - F(p_o)$. The value of partnership in Eq. (10) is the total expected value of market expansion and market contraction. ■

In order to prove Proposition 2, we need following technical Lemma.

LEMMA 4. *The value of partnership with online delivery platform $i \in \{\text{OM}, \text{OP}\}$ is positive, i.e., $VOP^i \geq 0$, when*

$$p_o^i \leq F^{-1}\left(\frac{\overbrace{1 - v^{catchment}}^{\text{Market expansion}}}{\underbrace{(1 - v^{catchment})}_{\text{Market expansion}} + \underbrace{(v^{catchment} - \overline{v}^i)}_{\text{Market contraction}}}\right), \quad i \in \{\text{OM}, \text{OP}\}. \tag{A.5}$$

Proof of Lemma 4. From value of partnership in Eq. (1) one can verify that $\text{VOP}^i \geq 0$ if

$$F(p_o^i) \leq \frac{1 - v^{\text{catchment}}}{(1 - v^{\text{catchment}}) + (v^{\text{catchment}} - \overline{v}^i)}, \; i \in \{\text{OM}, \text{OP}\}$$

This readily results in Eq. (A.5). ■

Proof of Proposition 2. Note from Lemma 4, the VOP increases when the right-hand side of Eq. (A.5) increases. But, the right-hand side of Eq. (A.5) can be further rewritten as $\frac{1-v^{\text{catchment}}}{1-\overline{v}^i}$, $i \in \{\text{OM}, \text{OP}\}$. Moreover, the term $v^{\text{catchment}}$ in the numerator originates from a fixed segment of customers who previously purchased from the outside option compared to BM store, independent of the partnership. Therefore, to study how VOP may change w.r.t different parameters related to online delivery platforms, we need to focus on the term $\overline{v}^i$ in the denominator. Specifically, the VOP increases when $\overline{v}^i$ increases, which is equivalent to reduction in market contraction effect. We are therefore interested in finding $\frac{\partial \overline{v}^i}{\partial r}$, $\frac{\partial \overline{v}^i}{\partial p_b}$, and $\frac{\partial \overline{v}^i}{\partial \sigma_p}$.

Consequently, we can define the following equations for membership platform $i = \text{OM}$:

$$\frac{\partial \overline{v}^{\text{OM}}}{\partial r} = \frac{\left(\sigma_p^2 - 2(\mu_p - 2p_b + p_o^{\text{OM}})\sigma_p + (\mu_p - p_o^{\text{OM}})^2\right)^2 r^2 - 16\phi^2\sigma_p^2}{32hr\sigma_p^2} \tag{A.6}$$

$$\frac{\partial \overline{v}^{\text{OM}}}{\partial p_b} = \frac{\left(\sigma_p^2 - 2(\mu_p - 2p_b + p_o^{\text{OM}})\sigma_p + (\mu_p - p_o^{\text{OM}})^2\right) r - 4\phi\sigma_p}{4h\sigma_p} \tag{A.7}$$

$$\frac{\partial \overline{v}^{\text{OM}}}{\partial \sigma_p} = \frac{\left((\mu_p - p_o^{\text{OM}})^2 - \sigma_p^2\right)\left(4\phi\sigma_p - \left(\sigma_p^2 - 2(\mu_p - 2p_b + p_o^{\text{OM}})\sigma_p + (\mu_p - p_o^{\text{OM}})^2\right) r\right)}{16h\sigma_p^3} \tag{A.8}$$

Similarly, we can define the followings for per-order platform $i = \text{OP}$:

$$\frac{\partial \overline{v}^{\text{OP}}}{\partial r} = \frac{\frac{-4\left(\sigma_p^2 - 2(\mu_p - 2p_b + p_o^{\text{OP}})\sigma_p + (\mu_p - p_o^{\text{OP}})^2\right)\sqrt{2}\sigma_p h\lambda}{\sqrt{h\lambda r}} + \left(\sigma_p^2 - 2(\mu_p - 2p_b + p_o^{\text{OP}})\sigma_p + (\mu_p - p_o^{\text{OP}})^2\right)^2}{32h\sigma_p^2} \tag{A.9}$$

$$\frac{\partial \overline{v}^{\text{OP}}}{\partial p_b} = \frac{-4\sqrt{2h\lambda r}\sigma_p + \left(\sigma_p^2 - 2(\mu_p - 2p_b + p_o^{\text{OP}})\sigma_p + (\mu_p - p_o^{\text{OP}})^2\right) r}{4h\sigma_p} \tag{A.10}$$

$$\frac{\partial \overline{v}^{\text{OP}}}{\partial \sigma_p} = \frac{\left(-4\sqrt{2h\lambda r}\sigma_p + (\sigma_p^2 - 2(\mu_p - 2p_b + p_o^{\text{OP}})\sigma_p + (\mu_p - p_o^{\text{OP}})^2) r\right)\left(\sigma_p^2 - (\mu_p - p_o^{\text{OP}})^2\right)}{16h\sigma_p^3} \tag{A.11}$$

To characterize the signs of the above expressions, we introduce the following notation for $i \in \{\text{OP}, \text{OM}\}$:

$$\Delta_i \equiv p_o^i - p_b, \qquad B_i \equiv \mu_p - p_o^i, \qquad A_i \equiv (\sigma_p - B)^2 - 4\sigma_p\Delta^i = \sigma_p^2 - 2(\mu_p - 2p_b + p_o^i)\sigma_p + (\mu_p - p_o^i)^2.$$

We can now rewrite the equivalents for the above equations. Specifically, the equations (A.6-A.8) for membership platform $i = \text{OM}$ can be rewritten as follows:

$$\frac{\partial \overline{v}^{\text{OM}}}{\partial r} = \frac{A^2 r^2 - 16\phi^2 \sigma_p^2}{32 h\, r\, \sigma_p^2}, \qquad \frac{\partial \overline{v}^{\text{OM}}}{\partial p_b} = \frac{A\, r - 4\phi\, \sigma_p}{4h\, \sigma_p}, \qquad \frac{\partial \overline{v}^{\text{OM}}}{\partial \sigma_p} = \frac{(B^2 - \sigma_p^2)\,(4\phi\, \sigma_p - A\, r)}{16 h\, \sigma_p^3}.$$

Furthermore, let $\Delta^{\text{OM}}_{max} \equiv \dfrac{(\sigma_p - B_{\text{OM}})^2}{4\sigma_p}$. We can then investigate two scenarios:

- $A_{\text{OM}} > 0$ or $\Delta_{\text{OM}} < \Delta^{\text{OM}}_{max}$: Let $\overline{r}^{\text{OM}} = \dfrac{4\phi\, \sigma_p}{A_{\text{OM}}}$. If $r \geq \overline{r}^{\text{OM}}$, then $\dfrac{\partial \overline{v}^{\text{OM}}}{\partial r} \geq 0$ and $\dfrac{\partial \overline{v}^{\text{OM}}}{\partial p_b} \geq 0$. Moreover, because $B^2_{\text{OM}} - \sigma_p^2 \leq 0$, then $\dfrac{\partial \overline{v}^{\text{OM}}}{\partial \sigma_p} \geq 0$ whenever $4\phi\sigma_p - A_{\text{OM}} r \leq 0$, or $r \geq \overline{r}^{\text{OM}}$. However, when $r < \overline{r}^{\text{OM}}$ we have $\dfrac{\partial \overline{v}^{\text{OM}}}{\partial r} < 0$, $\dfrac{\partial \overline{v}^{\text{OM}}}{\partial p_b} < 0$, and $\dfrac{\partial \overline{v}^{\text{OM}}}{\partial \sigma_p} < 0$.
- $A_{\text{OM}} \leq 0$ or $\Delta_{\text{OM}} \geq \Delta^{\text{OM}}_{max}$: First, $\dfrac{\partial \overline{v}^{\text{OM}}}{\partial r} \geq 0$ if $A^2_{\text{OM}} r^2 - 16\phi^2\sigma_p^2 \geq 0$, or equivalently, $r \geq \overline{r}^{\text{OM}}$. However, for $A_{\text{OM}} \leq 0$, we have $A_{\text{OM}} r - 4\phi\sigma_p \leq 0$, hence $\dfrac{\partial \overline{v}^{\text{OM}}}{\partial p_b} < 0$. Finally, because $B^2_{\text{OM}} - \sigma_p^2 \leq 0$, then for $A_{\text{OM}} \leq 0$ we have $4\phi\sigma_p - A_{\text{OM}} r \leq 0$, hence $\dfrac{\partial \overline{v}^{\text{OM}}}{\partial \sigma_p} < 0$. If $r < \overline{r}^{\text{OM}}$ we have $\dfrac{\partial \overline{v}^{\text{OM}}}{\partial r} < 0$, $\dfrac{\partial \overline{v}^{\text{OM}}}{\partial p_b} < 0$, and $\dfrac{\partial \overline{v}^{\text{OM}}}{\partial \sigma_p} < 0$.

Similarly, the equations (A.9-A.11) for per-order platform $i = \text{OP}$ can be rewritten as follows:

$$\frac{\partial \overline{v}^{\text{OP}}}{\partial r} = \frac{A^2 - 4\sqrt{2}\, A\, \sigma_p\, \sqrt{h\lambda}\, r^{-1/2}}{32 h\, \sigma_p^2}, \qquad \frac{\partial \overline{v}^{\text{OP}}}{\partial p_b} = \frac{A\, r - 4\sigma_p\, \sqrt{2h\lambda r}}{4h\, \sigma_p},$$

$$\frac{\partial \overline{v}^{\text{OP}}}{\partial \sigma_p} = \frac{\left(A\, r - 4\sigma_p\, \sqrt{2h\lambda r}\right)(\sigma_p^2 - B^2)}{16 h\, \sigma_p^3}.$$

With $\Delta^{\text{OP}}_{max} \equiv \dfrac{(\sigma_p - B_{\text{OP}})^2}{4\sigma_p}$, we can then investigate two scenarios:

- $A_{\text{OP}} > 0$ or $\Delta < \Delta_{max}$: Let $\overline{r}^{\text{OP}} = \dfrac{32\sigma_p^2 h\lambda}{A^2_{\text{OP}}}$. If $r \geq \overline{r}^{\text{OP}}$, then one can verify that $\dfrac{\partial \overline{v}^{\text{OP}}}{\partial r} \geq 0$ and $\dfrac{\partial \overline{v}^{\text{OP}}}{\partial p_b} \geq 0$. Moreover, because $\sigma_p^2 - B^2_{\text{OP}} \geq 0$, then $\dfrac{\partial \overline{v}^{\text{OP}}}{\partial \sigma_p} \geq 0$ whenever $A_{\text{OP}} r - 4\sigma_p \sqrt{2h\lambda r} \geq 0$, or $r \geq \overline{r}^{\text{OP}}$. If $r < \overline{r}^{\text{OP}}$ we have $\dfrac{\partial \overline{v}^{\text{OP}}}{\partial r} < 0$, $\dfrac{\partial \overline{v}^{\text{OP}}}{\partial p_b} < 0$, and $\dfrac{\partial \overline{v}^{\text{OP}}}{\partial \sigma_p} < 0$.
- $A_{\text{OP}} \leq 0$ or $\Delta \geq \Delta_{max}$: First, $\dfrac{\partial \overline{v}^{\text{OM}}}{\partial r} \geq 0$ if $A^2_{\text{OP}} - 4\sqrt{2}\, A_{\text{OP}}\, \sigma_p\, \sqrt{h\lambda}\, r^{-1/2} \geq 0$, which is met for any $r \geq 0$. However, for $A_{\text{OP}} \leq 0$, we have $A_{\text{OP}}\, r - 4\sigma_p\, \sqrt{2h\lambda r} \leq 0$, hence $\dfrac{\partial \overline{v}^{\text{OP}}}{\partial p_b} < 0$. Finally, because $\sigma_p^2 - B^2_{\text{OP}} \geq 0$, then for $A_{\text{OP}} \leq 0$ we have $A_{\text{OP}}\, r - 4\sigma_p\, \sqrt{2h\lambda r} \leq 0$, hence $\dfrac{\partial \overline{v}^{\text{OP}}}{\partial \sigma_p} < 0$.

Using $\Delta^i_{\text{max}} \equiv \frac{(\sigma_p - B)^2}{4\,\sigma_p}$ for $i \in \{\text{OP}, \text{OM}\}$, we can summarize the key insights:

– If $\Delta_i < \Delta^i_{\text{max}}$, or equivalently, $p_b \geq p^i_o - \frac{(\sigma_p - (\mu_p - p^i_o))^2}{4\sigma_p} \equiv \overline{p}^i_b$, then for each model $i \in \{\text{OM}, \text{OP}\}$ there exists a demand threshold $\overline{r}^i$ such that for all $r \geq \overline{r}^i$:

$$\frac{\partial \text{VOP}^i}{\partial r} \;\geq\; 0, \qquad \frac{\partial \text{VOP}^i}{\partial p_b} \;\geq\; 0, \qquad \frac{\partial \text{VOP}^i}{\partial \sigma_p} \;\geq\; 0, i \in \{\text{OM}, \text{OP}\}$$

– If $\Delta_i \geq \Delta^i_{\max}$, or equivalently, $p_b < p^i_o - \frac{(\sigma_p-(\mu_p-p^i_o))^2}{4\sigma_p} \equiv \bar{p}^i_b$, then

—For all $r > \overline{r}^i$:

$$\frac{\partial \mathrm{VOP}^i}{\partial r} \geq 0, \qquad \frac{\partial \mathrm{VOP}^i}{\partial p_b} < 0, \qquad \frac{\partial \mathrm{VOP}^i}{\partial \sigma_p} < 0, i \in \{\mathrm{OM}, \mathrm{OP}\}$$

—For all $r \leq \overline{r}^i$:

$$\frac{\partial \mathrm{VOP}^{\mathrm{OM}}}{\partial r} < 0, \qquad \frac{\partial \mathrm{VOP}^{\mathrm{OM}}}{\partial p_b} < 0, \qquad \frac{\partial \mathrm{VOP}^{\mathrm{OM}}}{\partial \sigma_p} < 0,$$

$$\frac{\partial \mathrm{VOP}^{\mathrm{OP}}}{\partial r} \geq 0, \qquad \frac{\partial \mathrm{VOP}^{\mathrm{OP}}}{\partial p_b} < 0, \qquad \frac{\partial \mathrm{VOP}^{\mathrm{OP}}}{\partial \sigma_p} < 0,$$

where

$$\overline{r}^{\mathrm{OM}} = \left[\frac{4\,\sigma_p}{(\sigma_p - B)^2 - 4\,\sigma_p\,\Delta}\right]\phi, \qquad \overline{r}^{\mathrm{OP}} = \left[\frac{32\,\sigma_p^2}{\left((\sigma_p - B)^2 - 4\,\sigma_p\,\Delta\right)^2}\right]\lambda.$$

Let $\overline{\overline{r}} = \max\{\overline{r}^{\mathrm{OM}}, \overline{r}^{\mathrm{OP}}\}$. For tractability, we assume grocery products are fast moving, with sufficiently high consumption rates (i.e., $r \geq \overline{\overline{r}} \triangleq \max\left\{\overline{r}^{\mathrm{OM}}, \overline{r}^{\mathrm{OP}}\right\}$). This aligns with the empirical context: groceries are purchased and replenished frequently (e.g., Canadian households average 1.8 baskets per week, with 86% shopping at least weekly (Flipp Market Research 2025)). We also assume online prices are higher on membership-based than pay-per-order platforms (i.e., $p^{\mathrm{OP}}_o \geq p^{\mathrm{OM}}_o$), reflecting industry practice where both platform types mark up in-store prices as platform revenue (Lampariello 2025), with membership platforms (e.g., Shipt) typically applying smaller premiums than pay-per-order platforms (e.g., Instacart), partially offsetting the upfront membership cost for customers (Lampariello 2026).

In the above analysis,

$$\frac{\partial \bar{p}^i_b}{\partial p_o} = \frac{\sigma_p + (\mu_p - p^i_o)}{2\sigma_p} > 0, \; i \in \{\mathrm{OM}, \mathrm{OP}\}$$

Because $p^{\mathrm{OP}}_o \geq p^{\mathrm{OM}}_o$, hence, it follows $\bar{p}^{\mathrm{OP}}_b \geq \bar{p}^{\mathrm{OM}}_b$. Figure A.2 and Table A.1 summarize these results. Finally, for Region A_2, the comparative-statics statements in Proposition 2 follow directly from the analysis above.

■

Proof of Proposition 3. Suppose VOP is positive for both partnership models. From Proposition 1, one can verify that VOP increases with $v^{\mathrm{BM}\to i}$, $i \in \{\mathrm{OM}, \mathrm{OP}\}$. Therefore, the difference function

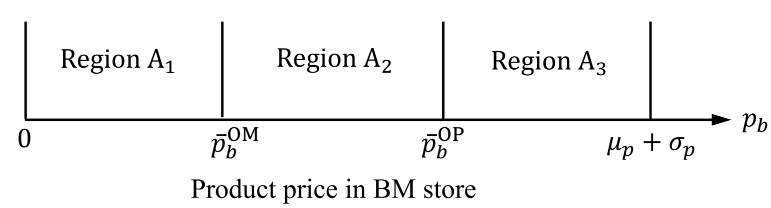


Figure A.2 Classification of the regions to study VOP

Table A.1 **Impact of price p_b, consumption rate r, and price dispersion σ_p on the VOP.**

Region	Platform	$\dfrac{\partial \text{VOP}^i}{\partial r}$	$\dfrac{\partial \text{VOP}^i}{\partial p_b}$	$\dfrac{\partial \text{VOP}^i}{\partial \sigma_p}$
$\mathcal{A}_1$	OM	>0	<0	<0
	OP	>0	<0	<0
$\mathcal{A}_2$	OM	>0	>0	>0
	OP	>0	<0	<0
$\mathcal{A}_3$	OM	>0	>0	>0
	OP	>0	>0	>0

$\Delta_{\text{VOP}}^{\text{OP}-\text{OM}}$, which shows $\text{VOP}^{\text{OP}} - \text{VOP}^{\text{OM}}$, increases when the function Δ_v increases, where $\Delta_v \equiv \overline{v}^{\text{OP}} - \overline{v}^{\text{OM}}$. In other words, to understand which partnership delivers more value–and how that preference shifts with key parameters–it suffices to study the slope of Δ_v with respect to the in-store price p_b, the consumption rate r, and the online dispersion σ_p.

For the simplicity in exposition, one can write $v^{\text{BM}\to i}$, $i \in \{\text{OM}, \text{OP}\}$ in the perfect-square forms:

$$\overline{v}^{\text{OP}} = \frac{(A\sqrt{r} - 4\sigma_p\sqrt{2h\lambda})^2}{32hr\sigma_p^2}, \qquad \overline{v}^{\text{OM}} = \frac{(Ar - 4\phi\sigma_p)^2}{32hr\sigma_p^2},$$

with $B \equiv \mu_p - p_o$, $\Delta \equiv p_o - p_b$ and $A \equiv (\sigma_p - B)^2 - 4\sigma_p\Delta$. Since $\sigma_p^2 > B^2$, the comparative statics of Δ_v become particularly transparent, as we discuss below.

We first begin with the in-store price p_b. Differentiating and simplifying shows that the sensitivity of the difference to p_b is $\partial\Delta_v/\partial p_b = (\phi - \sqrt{2h\lambda, r})/h$. This expression does not depend on A. Rather, the delivery fee per-order results in a greater burden for customers than the membership-fee, so higher p_b tilts the comparison toward the membership model and Δ_v falls with p_b.

Next we study the impact of online price dispersion σ_p. Using term derivatives and subtracting, one can obtain $\frac{\partial\Delta_v}{\partial\sigma_p} = \frac{\sigma_p^2 - B^2}{4h\sigma_p^2}\left(\phi - \sqrt{2h\lambda r}\right)$. Since $\sigma_p^2 \geq B^2$, the prefactor is strictly positive, so price dispersion affects Δ_v same as p_b. Specifically, increasing σ_p lowers Δ_v (i.e., membership becomes relatively more attractive).

Finally, consider consumption rate itself. Differentiating Δ_v with respect to r while holding A and σ_p fixed yields a rational expression with three terms:

$$\frac{\partial\Delta_v}{\partial r} = \frac{1}{32h\,\sigma_p^2}\left[A^2(1-r) - \frac{4A\sigma_p\sqrt{2h\lambda}}{\sqrt{r}} + \frac{16\phi^2\sigma_p^2}{r}\right].$$

Clearly, a negative term linear in r coming from the membership numerator $(Ar - 4\phi\sigma_p)^2$; a negative term proportional to $1/\sqrt{r}$ arising from per-order frictions $\sqrt{h\lambda r}$; and a positive term proportional to $1/r$ reflecting the fixed membership fee. The balance of these terms delivers a simple shape: for sufficiently small r, the $1/r$ term dominates and $\frac{\partial\Delta_v}{\partial r} > 0$; for sufficiently large r, the linear $-A^2r$

term dominates and $\frac{\partial \Delta_v}{\partial r} < 0$. By continuity, $\Delta_v(r)$ therefore increases at low consumption rate, reaches at least one interior maximum, and decreases at high demand. Economically, the pay-per-order model can outperform the membership-based model at very low purchase frequency (low consumption rate) because there is no fixed fee to amortize. However, as consumption rate scales, the membership-based model's value grows faster and ultimately dominates the pay-per-order model.

Taken together, the above analysis shows that the additional value the retailer derives from partnering with OP-platform relative to OM-platform (Δ_{VOP}) is decreasing in p_b, σ_p, and r. ■

B. Text-based Product Similarity Matching

The product descriptions and the hierarchical product taxonomy are different between the retailer data and Numerator data, with the exception of department labels (e.g., produce, dairy, meat, frozen food, etc.). To identify similar products between the two datasets, following the literature (Hwang et al. 2010, Nageswaran et al. 2025), we implement a text-based similarity matching procedure within each department.

First, for every product, we construct a text string by combining the product description with its category and subcategory labels. We preprocess this text by tokenizing it into individual words, applying stemming to reduce words to their root forms, and removing common stop words to ensure consistency in representation (Sarwar et al. 2001). Second, we construct a vocabulary across all products within a department and retain only tokens that appear with sufficient frequency, dropping rare tokens that are unlikely to aid matching. Each product is then represented as a vector over this vocabulary using a bag-of-words representation, where each element indicates the presence of a token in the product's text. Third, for each focal product i in the retailer data, we compute cosine similarity between its vector representation and that of every product in the Numerator data within the same department. Cosine similarity measures the degree of overlap in token usage between two products, scaled by the length of their respective vectors.

To illustrate, consider the retailer product "BE SF RICED CAULI" in the frozen food department under the category "FZ VEGETABLES" and subcategory "FZ STEAM VEGETABLES". After preprocessing and filtering rare tokens, this product is represented by the token set *(rice, veget, steam)*. A candidate products in the Numerator data–"KROGER RICED CAULIFLOWER", under category "Frozen Vegetables" and subcategory "Frozen Cauliflower"–is represented by *(kroger, rice, cauliflow, frozen, veget)*. The cosine similarity between these two products is determined by the number of shared tokens (i.e., *rice* and *veget*) relative to the geometric mean of their vector lengths.

$$\text{similarity} = \frac{A \cdot B}{||A|| \times ||B||} = \frac{2}{\sqrt{3} \times \sqrt{5}} = \frac{2}{\sqrt{15}} \approx 0.5164$$

Finally, we classify a Numerator product as a match to product i if the cosine similarity exceeds a threshold of 0.5. This threshold ensures sufficient textual overlap to identify comparable products while preserving enough variation across matched items for meaningful measurement of price dispersion.

C. Propensity Score Matching

We implement a two-step matching process. In the first step, we pair stores that begin using Shipt with those that do not adopt any platform. In the second step, we repeat this process for stores adopting Instacart. This approach enables us to isolate stores adopting Shipt or Instacart while ensuring comparability to non-adopting stores. For matching, we use store-level variables from our treatment assignment model and estimate propensity scores using logistic regression, following the methodology of Ertekin et al. (2024). We apply k-th nearest neighbor matching and adhere to a common support restriction, maintaining a tight caliper range of 0.01. Moreover, to mitigate potential spillover effects of treatment stores on control stores, we exclude control stores that are located in the same ZIPCode of the treatment stores. This methodology yields 168 matched stores without any platform during our sample period, 226 launched Instacart, and 42 launched Shipt. We define the 19-week period prior to the treatment week (the week of May 7th, 2018) as the prior-treatment period and the 20-week period post the treatment week along with the treatment week as the post-treatment period. In Table C.1, we present the balance diagnostics across covariates to assess the quality of matching between Shipt, Instacart, and non-platform stores.

Table C.1 Propensity Score Matching between No Partnership and Shipt and Instacart Partnerships

		Matching between No platforms and Shipt						Matching between No platforms and Instacart					
	Unmatched		Mean	Bias	% Reduction	t-test			Mean	Bias	% Reduction	t-test	
	Matched	Shipt	No platforms	%	$\lvert Bias \rvert$	t	p>$\lvert t \rvert$	Instacart	No platforms	%	$\lvert Bias \rvert$	t	p>$\lvert t \rvert$
Number of UPCs	U	9.5	9.418	47.4		2.65	0.008	9.568	9.419	-91.3		10.2	0
	M	9.497	9.495	1.2	97.4	0.06	0.953	9.522	9.513	-5.7	93.7	0.64	0.52
Average number of registers	U	2.399	2.206	67.7		4.21	0	2.218	2.209	-3.7		0.42	0.675
	M	2.38	2.351	10	85.3	0.48	0.634	2.296	2.27	-10.2	-171.6	0.93	0.355
Number of household	U	9.527	8.916	117		6.3	0	9.277	8.937	-62.7		7.04	0
	M	9.482	9.404	15	87.2	0.86	0.393	9.088	9.086	-0.4	99.4	0.04	0.97
Mean income	U	11.283	11.032	98.6		6.16	0	11.206	11.035	-63.9		7.23	0
	M	11.246	11.22	10.2	89.6	0.49	0.626	11.113	11.098	-5.4	91.6	0.52	0.607
Average price	U	1.395	1.353	58		3.01	0.003	1.415	1.353	-85.6		9.55	0
	M	1.391	1.39	1.3	97.7	0.06	0.952	1.399	1.39	-12.9	84.9	1.44	0.152

D. Counterfactual Analysis

To quantify counterfactual changes in sales at the SKU level, we estimate the following triple-difference model within a random-slope multilevel modeling framework (Rabe-Hesketh and Skrondal 2008) for the Shipt and Instacart partnerships separately. This specification allows the effect of platform partnership to vary across products.

$$
\begin{aligned}
Sales_{ist} = {} & \beta_0 + \beta_1 Treatment_{is} \times After_t + \beta_2 Treatment_{is} + \beta_3 After_t + \beta_4 Price_{is} + \beta_5 Consumption_{is} \\
& + \beta_6 Dispersion_i + \xi_{0i} + \xi_{1i} Shipt_{is} \times After_t + \tau_t + \varepsilon_{ist},
\end{aligned}
$$

In this specification, ξ_{0i} represents the deviation of SKU i's intercept from the overall intercept β_0, while ξ_{1i} captures SKU-specific deviations from the average partnership effect β_1. Thus, for each product i, the estimated sales effect of the partnership is given by $\beta_1 + \xi_{1i}$.

Estimation results are reported in Tables D.1 and D.2. For each estimation, we perform a likelihood-ratio test comparing the random-slope model to a corresponding ordinary linear regression. The test results support the inclusion of random slopes, indicating that the effects of both partnerships vary significantly across products. Finally, the estimated average partnership effects (β_1) are consistent with those reported in Tables 5 and 6, where Shipt and Instacart effects are estimated separately. This consistency lends further credibility to the multilevel model specification.

In each specification, we obtain predicted sales for each SKU. This step results in sales (i.e., $e^{Sales_{ist}^{O1}}$) that the partnership is expected to generate under the current practice. We then simulate the counterfactual scenario of removing from the platform the SKUs with a negative partnership effect by setting their value of interaction term (i.e., $Treatment_{is} \times After_t$) to 0. We then estimate the predicted sales for each SKU under this counterfactual scenario (i.e., $e^{Sales_{ist}^{O2}}$). Thus, the counterfactual impact of the proposed strategy on a given SKU can be estimated as $e^{Sales_{ist}^{O2}} - e^{Sales_{ist}^{O1}}$. We then calculate average difference as

$$
\left(\frac{1}{S \cdot T} \sum_{s=1}^{S} \sum_{t=1}^{T} e^{Sales_{ist}^{O2}} \right) - \left(\frac{1}{S \cdot T} \sum_{s=1}^{S} \sum_{t=1}^{T} e^{Sales_{ist}^{O1}} \right)
$$

.

E. Hybrid Platforms

Many grocery delivery platforms simultaneously offer both pay-per-order and membership-based pricing schemes. In this section, we extend our baseline model to examine how the presence of both options affects the market expansion and contraction forces. As established in the previous section, the key determinant of a consumer's preference between pay-per-order and membership models is her consumption rate. When the consumption rate is sufficiently low, consumers strictly prefer the pay-per-order model; conversely, when the consumption rate is sufficiently high, all consumers prefer the membership model—even when both options are available. To ensure that both schemes coexist in equilibrium and are selected by different consumers, we introduce heterogeneity in consumption rates. Specifically, we assume that individual consumption rates are uniformly distributed over a compact support $[\underline{r}, \overline{r}]$.

Table D.1 **Multilevel Model Estimation Results-Shipt**

VARIABLES	Estimate	Std. Err.
Shipt $\times$ After	0.016***	(0.006)
Shipt	-0.072***	(0.001)
After	-0.091***	(0.002)
$\ln$ Price	2.049***	(0.006)
$\ln$ Consumption	0.069***	(0.001)
$\ln$ PriceDisp^{Shipt}	-0.443***	(0.066)
Week FE	Yes	
Deviation$_{\text{Shipt}\times\text{After}}$	0.039***	(0.002)
Deviation$_{\text{con}}$	1.334***	(0.051)
LR test stat.	4,927,828.217	
Observations	7,176,667	

Standard errors in parentheses
*** p<0.001, ** p<0.01, * p<0.05, + p<0.1

Table D.2 **Multilevel Model Estimation Results-Instacart**

VARIABLES	Estimate	Std. Err.
Instacart $\times$ After	-0.032***	(0.004)
Instacart	-0.0002	(0.0006)
After	-0.045***	(0.002)
$\ln$ Price	1.846***	(0.005)
$\ln$ Consumption	0.073***	(0.0006)
$\ln$ PriceDisp^{Instacart}	-0.364***	(0.058)
Week FE	Yes	
Deviation$_{\text{Instacart}\times\text{After}}$	0.024***	(0.0009)
Deviation$_{\text{con}}$	1.112***	(0.042)
LR test stat.	9,572,241.589	
Observations	13,452,754	

Standard errors in parentheses
*** p<0.001, ** p<0.01, * p<0.05, + p<0.1

The remainder of the model remains unchanged. Following the same steps as in the baseline analysis, we can characterize the expected cost incurred by a consumer under each plan. If a consumer selects the membership model (OM), her expected cost is given by $\text{Cost}_{\text{OM}}(r) = \phi + E_p \min(p_o^{\text{OM}}, p)r$, whereas under the pay-per-order model (OP), her expected cost is $\text{Cost}_{\text{OP}}(r) = E_p \min(p_o^{\text{OP}}, p)r + \sqrt{2h\lambda r}$. Thus, a consumer chooses the membership option over pay-per-order if and only if $\text{Cost}_{\text{OM}}(r) \leq \text{Cost}_{\text{OP}}(r)$. Using the characterization in Proposition 3, this condition implies a threshold structure: consumers with sufficiently high consumption rates, i.e., $r \geq \frac{\phi^2}{2h\lambda}$, choose the membership model, whereas those with $r < \frac{\phi^2}{2h\lambda}$ opt for the pay-per-order model. Conditional on the plan choice, the value-of-partnership (VOP) expressions for each model remain unchanged. Integrating over the distribution of consumption rates, the value of the hybrid model is given by

$$\text{VOP}^{\text{HYB}} = \frac{1}{\overline{r} - \underline{r}} \int_{\underline{r}}^{\overline{r}} \Big(\mathbf{1}_{\{r \geq \phi^2/(2h\lambda)\}} \text{VOP}^{\text{OM}}(r) + \mathbf{1}_{\{r < \phi^2/(2h\lambda)\}} \text{VOP}^{\text{OP}}(r) \Big) dr, \tag{A.1}$$

where $\mathbf{1}_{\{\cdot\}}$ denotes the indicator function capturing the endogenous plan choice. The above analysis shows that introducing hybrid plan availability does *not* create a new economic channel. It simply splits the online segment into two sub-segments between model OM and model OP. Crucially, the object that drives all qualitative conclusions in the main model—the tension between market expansion and contraction—continues to drive outcomes in this hybrid setting as well. As a result, any comparative statics or welfare logic in the baseline that is built on the expansion-versus-contraction tradeoff carries over *plan-by-plan* in the hybrid environment.